AMAZON · L5 SUPPLY CHAIN MANAGER · TEAM MATCHING · ONE HOUR · TRANSPORTATION ENGINEERING SYSTEMS

Team-match conversation with Avinash Magima, Senior Manager, Transportation Engineering Systems

You. Rajesh Rishiyur Chandrasekaran. Offer in hand at L5 Supply Chain Manager; this call decides the team.

Him. Avinash Magima, Senior Manager, TESS, Bellevue. Industrial engineer (MS, Texas Tech), production-planning and OR-modeling background, Lean Six Sigma; Amazon since October 2017 as product manager, senior product manager, then manager of program management since September 2020. TESS configures and optimizes the outbound transportation network from fulfillment centers through sort centers and delivery stations: network design changes, package flows, pull times, sort configuration, config governance, and the automation and analytics behind them.

What the recruiter asked for. A brief overview of your experience, your questions about the team, and examples against three competencies: Analytical and Problem-Solving; Cross-Functional and Stakeholder Management; Ownership and Proactivity.

Weighting. Elementum and Flexport carry the hour. Sephora S/4HANA and Utah dual inventory are back-ups. Every number has a MATH table; say the inputs, then the result. Where an input is marked illustrative, say "modeled on Finance's rates" and do not quote the input as fact.

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Part 1 — Opening: who I am and why this team

1. Tell me about yourself — transportation lens

Sixty seconds. Chronological. One transportation number per stop. Stop when you finish the last sentence and let him talk.

SAY THIS — 158 WORDS

I have spent fifteen years making a network's plan match what moves. I started at Manhattan Associates on order-management programs: three distribution centers, a hundred stores, and ninety-nine percent fill against a ninety-five percent goal at Cyber Monday. Then MetricStream, as a program manager. At Elementum, 2017 to 2019, I took two hundred carrier lanes off the brochure and onto the measured lead-time tail, and safety stock followed that tail instead of a pad. On a separate onboarding I phased a thousand lanes by freight spend, not by lane count. At Flexport, 2021 to 2022, I ran Shipment Visibility across forwarders and containers we did not carry: seven customers to sixty-eight, and cost to serve from sixty-six cents to three cents a container. Since April 2023 at Sephora I publish one inventory number per node. TESS is that job at Amazon's scale: configure the network so a package moves at the lowest cost that still meets the promise.

Thirty-second cut. Use this if he has already read the resume and wants the transportation line only.

SAY THIS — 71 WORDS

Fifteen years. Elementum, 2017 to 2019: two hundred lanes on measured lead time, and the lanes that carried eighty percent of spend went first. Flexport, 2021 to 2022: shipment visibility from seven customers to sixty-eight, cost to serve from sixty-six cents to three cents a container. Sephora since April 2023: one inventory number per node. I configure a network so freight moves at the lowest cost that still meets the promise.

Years on record: Elementum 2017–2019, Flexport 2021–22 for the visibility program (resume span runs into early 2023; say 2021 to 2022 unless asked), Sephora April 2023 to present. Manhattan and MetricStream have no years in the packet. Fifteen years is the reconciled tenure. Do not invent a MetricStream client or dollar. The five-hundred-ninety-seven-lane figure is an earlier-record user anchor; the sixty-second version above does not say it. The thirty-second version says "eighty percent of spend" without the lane count. If he asks the count, use story 4.1 and label it.

2. Why Amazon — transportation lens

Sixty seconds. Specific to TESS. No internal system names.

SAY THIS — 200 WORDS

I want Transportation Engineering Systems because the work is the lane, the clock, and the cost. At Elementum a brochure lead time was a date nobody had measured. I replaced it with the distribution on the lane, and stock moved only where the tail justified it. The middle mile is that decision at network scale. A critical pull time is the clock a building has to hit so the sort and the delivery station still make the promise. At Flexport an exception with no owner became a ticket, and the ticket was the backlog I ranked by carrier and terminal. That is how I would troubleshoot a package flow and how I would govern a config change. I also take cost apart. Sixty-six cents a container was tickets, a platform cost, and volume, not a rate. Cost to serve per package is the same split. Two things I know from the outside. Amazon cited NielsenIQ in February 2023 at about 1.9 days from order to delivery. In 2023 Amazon split the US network into eight regions. I want to own a piece of that configuration, with a gate in front of the change and the cost on the other side.

Thirty-second cut.

SAY THIS — 75 WORDS

TESS configures the outbound network, from the fulfillment center to the sort center to the delivery station. I have done the small version. Measured lane lead time instead of a brochure. Exceptions treated as the backlog, not as noise. Cost per container taken apart into tickets and platform, not left as a rate. I want that job on the middle mile, on cost to serve per package, with a gate before a config goes live.

The 1.9-day figure is a public citation. Amazon posted it in February 2023, attributing it to NielsenIQ. The underlying writeups date the measurement to about July 2022. Say "Amazon cited NielsenIQ in February 2023," not "I measured 1.9 days." The eight regions are from Amazon's own 2023 description of the US network. Do not add other public operating statistics unless he asks and you have the page. Do not name any internal planning system.

Part 2 — Stories by competency, with the math and the solution approach

3. Competency map

Elementum and Flexport carry the primary and second slots. Sephora is the back-up only. Lead with the transportation story in each row. Do not add the one-and-a-half million of lane carrying cost to the two million of safety stock.

COMPETENCY	PRIMARY STORY (COMPANY)	SECOND STORY (COMPANY)	BACK-UP (SEPHORA)	LEAD METRIC	MATH BLOCK REF
Analytical and problem-solving	4.1 Measured lanes, safety stock, spend cut (Elementum)	4.3b Crux against 4PL, T-30 scored against T-0; then 4.3 cost per container (Flexport)	4.8 One published available-for-sale	99 percent lane accuracy; 63 against 25 became 27 against 25, then 90 against an 80 bar; $0.66 to $0.03	4.1, 4.3b, 4.3
Cross-functional and stakeholder management	4.5 Partner route and the reuse rule (Flexport)	4.3b The comparison that kept 4PL (Flexport); 4.1 spend share taken two levels up (Elementum)	4.9 Council dates through two go-lives	Value in month 6, not 18; about $5 million avoided, said once; a write-off reversed on evidence	4.5, 4.3b
Ownership and proactivity	4.2 Eight domains, the measurement, the alert (Elementum)	4.4 The weekly review nobody owned (Flexport)	4.8 Oversell to zero on a signed definition	Class-A forecast accuracy 70 to 85 percent; review accuracy 60 to 90 percent	4.2, 4.4

> If he asks for a failure, use 4.10 and do not spend 4.8 on it. If he asks what you would do on a promise network you have not operated, use 4.11 and say design case in the first sentence. Hold 4.6, 4.7, 4.8, and 4.9 until he asks for another example or for the current job.

4. Stories

4.1 Elementum — 200 carrier lanes on actual lead time; the spend cut; the shortage alert

Analytical. Two programs for one manufacturer. The lane release and the forecast release are different dollars. Never add them. "597 lanes, 80 percent of spend" is a user anchor: say "in my earlier record."

SITUATION/TASK

At Elementum, from 2017 to 2019, a six-site manufacturer planned two hundred shipping lanes on the carrier's brochure. The same customer wanted inbound visibility on a thousand lanes, and completeness sat at about sixty percent. I owned the measurement and the onboarding standard. The target was a lead time I could defend, and a first wave that carried the freight spend rather than the lane count.

ACTION

I used about a year of shipments per lane and set the plan to the measured ninety-ninth percentile, not the mean. On the worked lane the brochure said eighteen days, the mean was twenty-one, and ninety-nine percent had arrived by day thirty-four. Almost all of the old safety-stock variance was an unmeasured lead-time swing, so I replaced the formula with cover to that tail plus demand cover. Finance signed a two-sided rule: release stock where the measured spread sat inside the old assumption, and add stock where the tail was fatter. On the thousand-lane ask I ranked by spend and took two levels up the lanes that carried eighty percent of spend, in my earlier record, behind a ninety-five percent valid advance-ship-notice gate. The shortage alert fired when days of cover fell inside that lane's ninetieth-percentile lead time.

RESULTS

On the worked lane the buffer went from 12,937 units to 8,102, and late arrivals from seven in a hundred to one in a hundred. Across two hundred lanes the program released 580,000 units, about one and a half million dollars, one time, at ninety-nine percent lane accuracy. Completeness went from sixty to ninety-five percent.

MATH

Lead time into safety stock. Lane A is the packet's worked lane. Brochure, demand, and the eight-day swing are the loop packet's illustrative lane. The 580,000 units and about $1.5 million are the record pair. The old formula is z times the square root of lead time times demand variance, plus demand squared times lead-time variance.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Demand part = LT x sigma_d squared	18 x 150 x 150. Brochure 18 days and sigma_d 150 are illustrative lane inputs.	405,000
2	Lead-time part = d squared x sigma_LT squared	500 x 500 x 8 x 8. d = 500 units a day. sigma_LT = 8 days, the ERP default, never measured.	16,000,000
3	Combined = square root of the sum	square root of (405,000 + 16,000,000) = square root of 16,405,000	4,050 in the packet
4	Service factor = z x combined	2.33 x 4,050. z = 2.33 because the signed service was 99 percent. Toolkit label.	9,437 units. Exact product is 9,436.5; the packet prints 9,437.
5	Share that was a guess	16,000,000 / 16,405,000	97.5 percent of the variance was the unmeasured lead-time swing
6	Manual pad	7 days x 500 units a day. The week of pad is planner habit, illustrative.	3,500 units
7	Buffer before	9,437 + 3,500	12,937 units, 25.9 days of demand
8	Lead-time cover after	d x (p99 - mean) = 500 x (34 - 21). Ranked p99 = 34 days, mean = 21. Record shape of the example lane.	6,500 units
9	Demand cover after	z x sigma_d x square root of mean LT = 2.33 x 150 x square root of 21. Packet uses 4.583 under the root.	1,602 units
10	Buffer after	6,500 + 1,602. Pad = 0 because the plan holds 99 percent.	8,102 units
11	Release on this lane	12,937 - 8,102	4,835 units, 37.4 percent less, late arrivals 7 in 100 to 1 in 100

STEP	FORMULA	SUBSTITUTION	RESULT
12	The other direction, Lane B	Before 5,238 units. After = 200 x (58 - 30) + 2.33 x 80 x square root of 30 = 5,600 + 1,021. Illustrative second lane.	6,621 units, stock in of 1,383. A blanket cut would have been short.
13	Program net	Record pair: 580,000 units and about $1.5 million, one time, zero stockouts in the watch window. Implied unit value = 1,500,000 / 580,000.	$2.59 a unit, derived. 580,000 x 2.59 = $1,502,200, about $2,200 over the stated $1.5 million because 2.59 is rounded. Say about $1.5 million.
14	Roll-up that ties to the net	150 lanes x 4,500 = 675,000 out. 50 lanes x 1,900 = 95,000 in. Both counts are the loop packet's illustrative split.	675,000 - 95,000 = 580,000, which matches the record net. Label the split illustrative.

The eighty percent spend cut. The thousand-lane program across twenty-five European locations is record. The 597 and the eighty percent are a user anchor. Say "in my earlier record." Do not say four hundred plus lanes.

STEP	FORMULA	SUBSTITUTION	RESULT
1	S(k) = cumulative freight-spend share	Lanes sorted by annual freight spend. S(597) = 80 percent. User anchor.	597 of 1,000 lanes carry 80 percent of spend. Do those first.
2	Spend left behind	100 percent - 80 percent	20 percent of spend stays behind the same gate. Say "the remainder," not a lane count.
3	Completeness before	events received / events expected = 480 / 800. Volumes illustrative. The 60 percent is record.	60 percent
4	Completeness after	760 / 800. Volumes illustrative. The 95 percent is record.	95 percent
5	Cohort gate	valid advance ship notices in test, at or above 95 percent, before activation. The gate is record. One cohort failed it twice on a timestamp format. The twice-held detail is a user anchor.	Held both times, then activated.
6	Freight result, if asked	about 10 percent less freight cost, about $2 million in six months, on the onboarded lanes. User anchor. Implied base = 2,000,000 / 0.10.	About $20 million a year, derived, not a record. Do not add this $2 million to the $1.5 million lane release or to the $2 million safety-stock release in 4.2. Three different nouns.

Shortage alert tied to lane lead time. The spoken rule is days of cover at or under the lane's ninetieth-percentile lead time. The packet does not fire the alert by subtracting safety stock from a position. It also refuses to count unfirmed planned orders as supply. A raw "open PO" add would have been the supply lie. The worked rows below are illustrative and are not one SKU.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Days of cover = on-hand / daily demand	1,200 / 100. Illustrative. Lane p90 = 14 days, illustrative.	12 days. 12 is at or under 14, so the alert fires a lead time early.
2	Counter-case	Same on-hand, faster lane, p90 = 9 days.	12 is above 9. No alert. Three days of slack.
3	Safety stock on the toolkit lane	SS = z x d x sigma_L = 2.33 x 100 x 4. Illustrative. This is not the Lane A formula in the table above.	932 units. If sigma_L falls from 4 days to 2, SS = 2.33 x 100 x 2 = 466. Release 466 units only if forecast error and lead-time error both improved.
4	Position = on-hand + in-transit that has landed + released orders - demand	Worked finished-good row, illustrative, a different SKU. Day 3: on-hand 400, in-transit has not landed, production has not landed, demand 500.	400 - 500 = -100. Below any positive safety stock. The shortage is dated.
5	Same row, day 7	(400 + 600) - (500 + 700). Day-9 production is excluded because it has not landed.	1,000 - 1,200 = -200. Fix window = day 9 - day 7 = 2 days.
6	Same row, day 10	(400 + 600 + 800) - (500 + 700 + 400)	1,800 - 1,600 = +200. Timing shortage, not a volume shortage.
7	What is not in the supply	Unfirmed planned orders are excluded on purpose. Record rule.	Not on record: one row that is on-hand + in-transit + every open PO - demand, compared with safety stock, for a live SKU. Do not build that row in the room.

SOLUTION APPROACH

Inputs were the lane history, daily demand and its spread, the brochure, and the ERP's default lead-time swing. The model was a variance split, then a two-term buffer that does not assume the lead-time tail is a bell curve. The decision rule was two-sided and signed: release where the measured spread and the tail sit inside the old assumption and lane accuracy holds; keep or add stock where they do not. I simplified the formula a planner had to defend, and I extended the onboarding by a published data gate and a portal path so a small carrier was not a custom project. The long-term risk I wrote down was a silent re-plan: a drift flag reopens the lane if the rolling tail moves more than two days past plan, the spread rises by a quarter, or four or more in a hundred run late.

IF THEY PROBE

- Why not promise the average? A promise at the mean is late about half the time on a skewed lane. The tail is the commitment. The ninetieth percentile is only the alert, so the planner still has a window.
- What would you do differently? Put the lane that needed more stock on the first page. The add-stock case is what made Finance trust the release.
- What did it cost? On the fat-tail lane, stock went in, 5,238 units to 6,621, illustrative. On the program, about twenty percent of spend waited behind the same gate. I also gave up the hand pad, which some planners treated as their own policy.
- How did Finance trust the million and a half? Lane by lane, the rule written before any stock moved, including the add-stock clause. It is one-time carrying cost released. I do not annualize it.
- Is 597 what went live? Apple and the resume say a thousand lanes across twenty-five European locations were onboarded. My earlier record says 597 first for eighty percent of spend. I say it that way, and I do not say four hundred plus lanes.
- Why did one cohort wait? It missed the ninety-five percent advance-ship-notice gate twice on a timestamp format. A bad date in a visibility product is the first thing the customer sees. Held twice, then activated.

4.2 Elementum — eight SAP domains into one supply view; class-A forecast accuracy 70 to 85 percent

Analytical and ownership. The eight domains belong to the six-site manufacturer. The contract manufacturer supplied the bill-of-materials feed. You owned the definitions and the measurement. You did not own the forecast.

SITUATION/TASK

At Elementum, 2017 to 2019, planners at that six-site manufacturer learned about a component shortage when the line stopped. Eight SAP domains never met in one supply view: supplier, material, bill of materials, inventory, production, transportation, sales order, and planning. I owned the definitions, the data-quality gates, and the way accuracy was measured. The client's SAP teams and our engineers built the feeds. The target was a shortage visible a lead time early, and a forecast accuracy number a planner could audit.

ACTION

I wrote the grain first: SKU, location, and date. Supply is on-hand, in-transit that will land, production that will land, and released orders only. Unfirmed orders stayed out, because they change every planning run. Each domain went live behind a gate its owner could fail. The feed was a nightly full load plus intraday deltas. A failed row went to quarantine and was fixed at the source. Morning numbers were reconciled to SAP before an alert was trusted. Accuracy was one minus mean absolute percentage error on class-A SKUs, with bias published beside it from month two. Nobody was ordered off a spreadsheet.

RESULTS

Class-A forecast accuracy went from 70 percent to 85 percent. The shortage alert was live in three months. Planner collaboration went up fifty percent once the side spreadsheets were no longer the system of record. About two million dollars of safety stock came out, one time, only where forecast error and lead-time error had both improved, under a rule Finance signed. I never add that two million to the lane program's million and a half.

MATH

The packet measures accuracy as one minus MAPE and publishes bias beside it. It does not give a weighted absolute percentage error. I will not invent one. The four-week sample below shows the arithmetic. It lands at 18.3 percent error. It does not reproduce the record 30 percent or 15 percent. Those two error rates are the record, stated, not derived from this sample.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Accuracy = 1 - MAPE	Before: MAPE = 0.30 on scoped class-A SKUs. Record.	1 - 0.30 = 70 percent
2	Accuracy after	MAPE = 0.15 on the same scope. Record. The fifteen-point move is the record, not a sample I can rebuild.	1 - 0.15 = 85 percent

STEP	FORMULA	SUBSTITUTION	RESULT
3	What MAPE is	Mean of absolute percent errors. Direction ignored. Bias is the signed mean error, published from month two so a chronic over-forecast cannot hide inside a decent MAPE.	MAPE steered the accuracy line. Bias had to move too before stock came out.
4	Illustrative four weeks, not the record	Actuals 100, 120, 80, 100. Forecasts 130, 110, 100, 90. Absolute errors 30, 10, 20, 10. MAPE = (30/100 + 10/120 + 20/80 + 10/100) / 4.	(0.30 + 0.083 + 0.25 + 0.10) / 4 = 18.3 percent. This sample is not the 70 or the 85.
5	Bias on that sample	(sum of forecasts - sum of actuals) / sum of actuals = (430 - 400) / 400	+7.5 percent. Over-forecast. Illustrative.
6	Why bias still matters at a 15 percent MAPE	A +10 percent over-forecast on 100 units a day adds 70 units a week. 932 / 70.	13.3 weeks and the excess equals the whole 932-unit buffer on the toolkit lane. Illustrative.
7	The two-million release	Record, one time, only where both errors improved. Not annual. The recurring pieces around it, carry at 10 percent, expedites, planner time, are a model of about $0.8 million a year. Say modeled if asked. Do not add to the $1.5 million.	$2 million on the record. The $0.8 million is not.
8	What is not on record	The weekly actuals and forecasts that produce exactly 30 percent MAPE and then 15 percent. A WAPE formula. A count of class-A SKUs.	Say the definition and the two endpoints. Do not reconstruct a history.

SOLUTION APPROACH

Inputs were the eight domains, the SAP lists I reconciled to, and the lane lead time from the measurement program. The model was an honest supply equation plus MAPE and bias at the family grain. The decision rule was: no alert on a day the reconciliation fails, and no safety stock out unless both error signals improved. I simplified by refusing unfirmed orders and by freezing the grain. I extended by a nightly full load plus intraday deltas, and by quarantine instead of a silent fix. The long-term risk was a planner who learns to ignore alerts. Three weeks of reconciliation before the first trusted alert was the mitigation. The part-number map slipped; I would start that worklist on day one.

IF THEY PROBE

- Did you own the forecast? No. Planners owned it. I owned the measurement and the input integrity. I will not say I ran demand planning end to end.
- What would you do differently? Start the part-number mapping worklist on day one. It gated longer than I planned, and I took the new date to the sponsor with the coverage number attached.
- What did it cost? Three months to a trusted alert, not three weeks to a screen. The two million is one-time working capital, not an annual saving. My model of the recurring value is separate and labeled.
- Why exclude unfirmed orders? They breathe every planning run. Counting them inflates supply, which is how the line-down surprises happened.
- Planner effort or planner collaboration? The record I will say is collaboration up fifty percent. I will not say effort down fifty percent. Those are different sentences, and only one is the reconciled line.
- Where does the contract manufacturer sit? They supplied the bill-of-materials feed. The eight domains are the manufacturer's. Overlapping revisions were quarantined for a person to disposition. I do not tell a bill-of-materials explosion story.

4.3 Flexport — Shipment Visibility: gates, one pipeline, cost per container

Cross-functional and analytical. Speak $0.66 to $0.03. Do not open with an automation percentage. Two worksheets in the packet use different automation rates to talk about the same cents. The note under the math says which line to use if he asks.

SITUATION/TASK

At Flexport, 2021 to 2022, shippers wanted one view of containers, including containers Flexport did not carry. When the service could not start tracking, it opened a ticket and a person typed the milestones. Seven pilot customers. Cost to serve was sixty-six cents a container. System accuracy was about sixty percent. I was the program owner across nine teams and about a hundred people. I did not own the engineering build. The target was a lower cost to serve while the book grew, and a customer graduated only when the gates held at the current load.

ACTION

I made the headline the share of containers tracked with no human touch. Every ticket was a failed automation, tagged by carrier and terminal, and we ranked the backlog by tickets times cost. Customers boarded in waves on six published gates, measured on their own containers. I held wave three when arrival accuracy was 76 percent against an 80 percent bar, fixed the source, and

retested for a week. Four systems became one pipeline only after a parallel run. A slow container stayed in the denominator for fifteen days past terminal exit, and I discarded a week that printed 99.4 percent on three touches.

RESULTS

Customers went from seven to sixty-eight, two months early. System accuracy went from 60 to 90 percent. The parallel run held within 1 percent before anyone switched. Cost to serve went from $0.66 to $0.03 a container, about a 95 percent cut, while volume grew about five times. Zero rollbacks.

MATH

Cost to serve, before. Packet page 1331. The two rates in brackets on that page are illustrative, chosen to land on Finance's real endpoint of $0.66. The container count, the 60 percent, and the 800 tickets are the worksheet's baseline for seven customers.

> **Do not say the minutes line.** The same page prints "12 min x $75 an hour = $1.50." That product is $15.00, not $1.50. Twelve minutes is 0.2 hours, and 0.2 times 75 is 15. The page is wrong by a factor of ten. Speak $1.50 as an illustrative rate chosen so the worksheet lands on $0.66. Do not derive it from minutes.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Tickets = containers x (1 - automation)	2,000 a week x (1 - 0.60). Worksheet baseline, seven customers.	800 tickets a week
2	Human cost = tickets x cost per ticket	800 x $1.50. The $1.50 is illustrative, chosen to land on the real endpoint. It is not 12 minutes times $75.	$1,200 a week
3	Platform cost	$120 a week, illustrative, same page.	$120 a week
4	Cost per container = (human + platform) / containers	(1,200 + 120) / 2,000 = 1,320 / 2,000	$0.66 exactly. This is the chain to say.
5	The minutes line, shown so you can see the error, not so you can say it	12 / 60 x 75	$15.00. The page prints $1.50. Do not say this row.

Cost to serve, after. Same page. These inputs do not reproduce $0.03 by themselves.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Volume	10,000 containers a week. The page calls this 5 times the 2,000 baseline. 68 customers.	10,000 a week
2	Tickets = containers x (1 - automation)	10,000 x (1 - 0.97)	300 tickets a week. The page states 300.
3	Human cost	300 x $0.80. The $0.80 is illustrative: a pre-filled carrier page and lookup tooling.	$240 a week
4	Human cost per container	240 / 10,000	$0.024
5	Platform cost per container	$150 a week, illustrative, divided by 10,000	$0.015
6	Sum of the two inputs on the page	0.024 + 0.015	$0.039. The page rounds this to about $0.04.
7	The stated endpoint	The page then writes "about $0.04 to $0.03 with bulk source pricing." It does not give the bulk price.	Gap = 0.039 - 0.030 = $0.009 a container. On 10,000 containers that is $90 a week of bulk-source discount that is not on the page. Do not invent the bulk rate. Say the worksheet lands at about four cents, and the three cents needs a bulk-pricing term the page does not quantify.
8	The cut you say aloud	1 - (0.03 / 0.66) = 1 - 0.04545	95.45 percent. Say "about ninety-five percent." The endpoints $0.66 and $0.03 are the record. The path between them is the worksheet above.

$0.18 and the 3.7 touches. Packet page 1237. This is a cousin check, not the ticket math.

STEP	FORMULA	SUBSTITUTION	RESULT
1	What $0.18 is	The platform priced one eliminated interaction, one touch, at $0.18. Finance's rate. It is not the $1.50 ticket and not a measured touch count.	$0.18 per eliminated interaction
2	Residual touches before = cost per container / price per touch	0.66 / 0.18	3.666..., which the page prints as about 3.7 touches a container
3	Residual touches after	0.03 / 0.18	0.166..., which the page prints as about 0.17 touches

Automation, if he asks. Do not blend these. The reconciled line from the Apple record is about 40 percent automated to about 90 percent, with a 70 percent no-touch gate. Page 1331's cost worksheet uses 60 percent to 97 percent because those rates produce 800 tickets and then 300 tickets and land on the cents. A different page decomposes the 22-times drop as manual share from 60 percent to 10 percent, which is the 40-to-90 line, times about 3.7 fewer touches. Say the cents. If he asks the rate, say the review record is about forty to ninety, and show the worksheet if he wants the cents built up. Do not say both rates as one measurement.

Graduation gates and the accuracy definition. Gates are record, from the pilot schedule. Accuracy here is system accuracy, not arrival accuracy.

STEP	FORMULA	SUBSTITUTION	RESULT
1	System accuracy	milestones matching the source of truth / milestones audited	60 percent to 90 percent. Record. Not an arrival-time accuracy.
2	Parallel-run gate	absolute value of (manual - pipeline), each frozen metric	At or under 1 percent for consecutive weeks, every mismatch explained, then switch.
3	Verification SLA	automated review within 24 hours, manual within 24 hours, nothing in progress beyond 48 hours	At or above 90 percent of the time
4	Internal data quality	against the source, and at least as accurate as the legacy product	At or above 90 percent
5	Milestone and arrival accuracy	separate from system accuracy	At or above 80 percent. Wave three was held at 76 against 80.
6	Engagement and stability	at least 100 containers and the key features, production stable, no show-stoppers, satisfaction at least the consignee average	All six gates, at the current load, before a wave boards. Entry bar about 50 containers so the product is calibrated before login.
7	Customer multiple	68 / 7	9.71, say 9.7 times. Do not round it to a clean ten.

SOLUTION APPROACH

Inputs were submitted containers, tickets, milestones, and a denominator that kept slow containers. The model was cost per container equals human cost plus platform cost, divided by containers, with the ticket as the sensor for what to automate next. The decision rule was a wave boards only when all six gates pass at the current load, and a source leaves the human loop only after it holds at or above 95 percent automated for four weeks, and re-enters below 90 percent. I simplified by freezing seven definitions and by refusing a public tracking site until cost and quality were fixed. I extended by keeping a human in the loop on purpose, with a reason code, instead of pretending automation was done. The long-term risk was cost per container rising as dirty feeds arrived. The gates are what kept unit cost falling while the book grew.

IF THEY PROBE

- What would you do differently? Put the security review on the kickoff list. Ticket data carries customer content. I found that gap mid-build. Story 4.10 is that miss, if he wants it in full.
- What did it cost? The manual path stayed until the parallel run passed, on the order of the weekly snapshot hours in story 4.4. Holding wave three cost a week of retest. I did not buy a competitor's visibility product. We kept buying raw carrier and terminal sources, because visibility was the product.
- What was yours? The definitions, the denominator, the gates, the ranking, and the cutover call. Engineering built the service.
- Why throw out a 99.4 percent week? Three touches. A flattering rate on three observations will steer a rollout. I printed the gap.

- Is 32,000 shipments a result? No. It was a public target I managed toward. Sixty-eight customers and 170 monthly active users were the targets that landed, two months early.
- How did volume alone not get you to three cents? The page says volume alone, spreading a fixed cost five times, would have left about thirteen cents. The rest is the automation. Say that only with the worksheet in front of you, and do not pretend the after-state inputs hit three cents without the unpriced bulk term.

4.3b Flexport — Crux against the 4PL platform: T-30 scored against T-0 on every milestone, and the comparison that kept 4PL

Analytical and stakeholder. This is the decision story inside 4.3. Use it when he asks how you changed a leadership decision with data, how you compare two systems, or how you handle a number that is about to kill a product. Say the normalized pair first, then the residual you named against yourself.

> Wording. Your record carries this comparison two ways: "Crux's 63 percent" and "a competitor's published 63 percent." You told me it was Crux. Say Crux, and say it the same way all hour. If he has read the record and asks, say "the legacy dashboard" and move on. 4PL means containers Flexport did not forward; 3PL means containers Flexport did forward and is the control group in the review.

SITUATION/TASK

At Flexport in 2022, after Shipment Visibility launched, leadership put two numbers side by side: the new 4PL platform at 25 percent schedule reliability, and Crux, the acquired tracker we still ran for 79 customers, at 63 percent. The question was whether to deprecate 4PL and keep Crux. The two numbers used different containers, milestones and scoring windows. I owned the retirement risk, the comparison and the evidence for the forum. The target was one definition both products were scored on before anyone decided, with the real gap named.

ACTION

I rebuilt the comparison instead of arguing it. One definition: for every container and every milestone, take the time predicted thirty days out and score it against the actual at time zero, inside one tolerance, on the same containers, sources and week. Crux's 63 had been scored on the milestones its feeds covered, close to the event. On the common basis it re-scored to about 27 against our 25. I changed the basis, not the team: their number stayed valid on its own denominator. I named our residual against ourselves, arrival variance of sixteen days against eight, and secured six months of source additions with gates. Decommission went to the top of the risk register, and all 79 Crux customers got a destination and a date.

RESULTS

The write-off died on evidence. 4PL stayed. Reliability went from the mid-twenties to 90 percent against an 80 percent north star once sources were added. Crux retired by segment, cancellations dated through October 24, 2022, and 294,581 annual containers each with a path. The rule that survived: reconcile before you compare.

MATH

How one container is scored. The horizon is the point before the event at which you freeze the prediction. The weekly review carried the instrument as accuracy at ATD minus 28, minus 10 and minus 5 days, with mean absolute error and the standard deviation of that error at each horizon (record-from-docs, LSA WBR 2022, Ocean FCL lines 30 to 33). Say T-30 as the round number; the sheet used T-28. Dates and the one-day tolerance below are illustrative.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Error at horizon h = predicted at T-h minus actual at T-0, in days	Arrival at port of discharge. Predicted at T-30: March 30. Actual: April 2.	3 days early. Illustrative container.
2	Hit if the absolute error is at or inside the tolerance	3 days against a tolerance of 1 day. Tolerance is illustrative; the record does not print it.	Miss at T-30
3	Same milestone, later horizons	At T-10 the prediction was April 1: error 1 day. At T-5 it was April 2: error 0.	Hit at T-10. Hit at T-5. The same container scores differently depending on when you freeze it.
4	Repeat for every milestone on the container	Departure estimate, actual departure, transshipment arrival, discharge-port arrival, discharge, gate-out, delivery. Each has its own error at each horizon.	One row per container per milestone per horizon
5	Reliability at horizon h = hits at h / milestones scored at h	Reported at T-28, T-10 and T-5 so the decay is visible.	Reliability rises as the horizon shortens. Two numbers at different horizons are not comparable.

Why 63 became about 27. The 63, the about 27 and the 25 are record. The populations below are illustrative and chosen so the division lands on the record figures.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Crux dashboard, its own basis	Scored only the container-milestones its carrier feeds covered, near the event: 400 of 1,000 in the week. Hits 252. 252 / 400.	63 percent. Valid on its own denominator.
2	Common basis for Crux	All 1,000 container-milestones, same week, same sources, frozen at T-30. Hits 270. 270 / 1,000.	About 27 percent
3	Common basis for the 4PL platform	Same 1,000. Hits 250. 250 / 1,000.	25 percent
4	Raw gap and normalized gap	63 minus 25. Then 27 minus 25.	38 points raw. 2 points normalized. 36 of the 38 points were basis, not product.
5	What the 2 points meant	Inside the week-to-week movement of either number.	No product decision could rest on it. The decision moved to the residual.

The residual I named against ourselves. Record figures.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Spread ratio = our variance / theirs	16 days / 8 days	2.0 times the spread. A T-30 prediction with twice the spread lands inside the same tolerance less often. That was the real gap.
2	What closes it	Sources, not arithmetic: direct carrier feeds and terminal feeds on the lanes we were weakest on.	Six months of source additions, with gates. Record.
3	Outcome against the bar	90 minus 80	10 points over the north star. Record. Real work, not normalization.

What was on the table. Record-from-docs, the Crux transition RAPID.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Annual recurring revenue = monthly x 12	$15,491 x 12	$185,892. Ties to the record.
2	Customers by segment	2 global key accounts + 14 enterprise + 39 SMB = 55 trade. 9 motor carriers + 3 truck brokers + 5 forwarders + 1 customs broker + 4 3PLs + 2 SaaS = 24 non-trade. 55 + 24.	79 customers
3	Revenue that could not move to the client app	About 75 percent of Crux revenue was non-trade. 0.75 x 185,892.	About $139,000 a year. Derived. Say "about three quarters of the revenue."
4	Containers a week	294,581 / 52	About 5,665 a week. Derived.
5	Cost to keep Crux running	Not on record.	Say "two platforms, two cost bases, two versions of the truth." Do not quote a dollar.

What the weekly review priced. Record-from-docs from the uploaded LSA WBR 2022 and the E-Team takeaways of February 1, 2022. This is where the $0.18 in story 4.3 comes from.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Price per interaction = cost-to-serve reduction / interactions eliminated	Forwarding Platform lines 2 and 12, week 48: $44,074.08 / 244,856. Week 49: $41,015.52 / 227,864.	$0.18 both weeks. Finance's price of one eliminated touch. It is the same $0.18 that makes 0.66 / 0.18 about 3.7 touches.
2	The E-Team decision, February 1, 2022	Goal 1: lower cost to serve by replacing human track-and-trace with the automated product, which also makes 4PL cheap to offer. Action: measure cost to serve for 3PL plus 4PL against 4PL only, in the weekly review.	The 4PL question was framed as a cost-to-serve question, not a feature question. That is the frame the comparison had to answer.

STEP	FORMULA	SUBSTITUTION	RESULT
3	The review the team published	As of February 14, 2022: the visibility team publishes its own weekly review; priority is arrival-estimate accuracy at the discharge port and interaction-elimination reporting.	Those two lines are the instrument that the T-30 scoring and the $0.18 fed.
4	Review rules	Five consecutive weeks of trend; green or red at a favorable or unfavorable variance above 5 percent relative; input metrics the team controls, output metrics the stakeholder cares about.	The plus or minus 5 percent week-over-week rule in your record is this sheet's rule.

SOLUTION APPROACH

Inputs were every milestone prediction the two platforms had held, with its timestamp, and the actuals. The model fixed three things before any division: the same containers, the same sources and week, and the same horizon, T-30, with the tolerance per milestone written down. The decision rule was reconcile before compare: no decision on two numbers with different denominators. I simplified by refusing a feature-by-feature debate and scoring one definition. I extended by reporting three horizons so the decay was visible, and by naming our own residual in the same readout. The long-term risk was a product written off on a measurement artifact, and a legacy that never retires because nobody owned the retirement. The risk-register entry, the per-segment destinations and the one forum that signed improvement and retirement in the same week are what closed it.

IF THEY PROBE

- Isn't normalizing just making your number look better? A normalization that only flatters is spin. This one exposed our variance at sixteen days against eight, and I led the readout with it.
- Why T-30 and not T-5? A shipper plans labor, drayage and inventory against the date it had a month out. At T-5 the ship is nearly in port and every tracker looks good. We reported T-28, T-10 and T-5 so nobody could pick the flattering one.
- Who decided? The product leads recommended and performed; two executives agreed and decided. I was input and I owned the transition mechanics. I will not say I was the decider.
- Why not keep Crux as well? Two platforms, two cost bases, two truths. About three quarters of Crux revenue was non-trade customers who could not use the client app, so each got a path: migrate, partner, API, or cancel with notice.
- What did it cost? $185,892 of annual recurring revenue retired. Six months of source work. The credibility risk of normalizing first, which the named residual protected.
- What would you do differently? Write the canonical-source rule before the first executive comparison. Once a dashboard has a constituency, a cheap definition becomes an expensive negotiation.
- Was Crux off on time? The target was end of 2022. The record shows dated cancellations through October 24, 2022, and the remaining trade clients on the second-half roll-out. I will not claim a single switch-off date that is not on record.

4.4 Flexport — weekly review automation, seven definitions

Ownership. This is the measurement system inside 4.3, told only if he asks how the review itself got built. Do not take credit for the customer count twice.

SITUATION/TASK

The Monday review for that same program was four exports I stitched by hand: tickets, the container pull, platform tracking, and blocker counts. Tickets had no container key. Accuracy was about sixty percent, and a vice president was steering a rollout on a number one person could break. I owned the definitions and the acceptance gate. Infrastructure built the pipeline. The target was the same seven numbers, every week, from one pipeline, with every discrepancy explained.

ACTION

I froze the seven definitions with the owners who would be red against them, and I fixed the denominator before anyone wrote a feed. Tickets had no container key, so we shipped a cross-reference and pushed a real key upstream. Old and new ran in parallel. Most mismatches were late tickets on the manual side. Nothing switched until every frozen metric matched within one percent for consecutive weeks. I priced the status quo in the requirements document, at the document's own assumed rate, so the infrastructure team could prioritize it without an escalation. Scope stayed at the seven metrics already in the review. When a routine update dropped about four percent of the ticket sync, the new alert caught it. The manual pull stayed as rollback until the gate passed.

RESULTS

Review accuracy went from 60 percent to 90 percent. Parity held within 1 percent. The first automated catch was that four percent sync drop. No Monday was skipped. The requirements document priced the old process at $103,680 a year on its assumptions. That figure is priced, not a saving I booked.

MATH

Hours. Two different manual-effort measures exist. Do not stack them. The requirements document prices three people across two reviews. Separately, a snapshot took 1.5 to 2 hours, and your own account of the scorecard is about four hours a week at its worst. The dollar line uses the three-person load and an assumed $90 an hour.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Hours before = people x hours x reviews	3 x 5 x 2. Document arithmetic. The $90 an hour is assumed, not a measured wage.	30 hours a week
2	Hours after	1 hour a week per person per review: 3 x 1 x 2	6 hours a week
3	Hours saved a week	30 - 6	24 hours a week
4	Annual priced baseline, before	30 x 48 weeks x $90. The document also prints 30 x $90 x 4 weeks = $10,800 a month. 10,800 x 12 = 129,600. Same 48-week year. A printed monthly of $10,880 is a typo. Say $10,800 if you quote a month.	$129,600 a year, priced
5	Annual after	6 x 48 x $90	$25,920 a year, priced
6	Difference	129,600 - 25,920, which is also (30 - 6) x 48 x 90	$103,680 a year. Priced, not a claimed saving.
7	The other effort figures, not added	1.5 to 2 hours per weekly snapshot, document as-is. About four hours a week at its worst, your account of the scorecard.	Say one of them and name which. Not on record: a single timesheet that reconciles 1.5 hours, 4 hours, and 30 hours. They are different grains.

SOLUTION APPROACH

Inputs were four exports and seven definitions that already existed. The model was a container-week grain and a join from ticket to container, with week-over-week as the relative change and a red or green beyond five percent. The decision rule was parity within one percent for consecutive weeks, every miss explained, or no cutover. I simplified by refusing new metrics in the first pipeline. I extended by an alert on a failed feed, so silence was not a green week. The long-term risk was a single person who could break Monday. The pipeline and a second owner are what removed that. The security lead time was the risk I missed, in 4.10.

IF THEY PROBE

- What would you do differently? Schedule the access review at kickoff, with its lead time, not when the feed is ready.
- What did it cost? Two extra weeks on the manual path when that review surfaced, and the ongoing six hours a week, not zero. The $103,680 is the price of the old process, not money saved.
- Name the seven metrics. The document says it names seven and that I froze the ones already in the review. I will stand on cost per container, system accuracy, automation rate, and the forwarder-versus-carrier timeliness pair. I will not invent the other names.
- Why only ninety percent accuracy if the gate was one percent? Ninety percent is the product measure, milestones against the source. One percent is the migration gate, new pipeline against the manual pull. Different questions.
- Who disagreed? Nobody had to. Infrastructure prioritized it because the price was in their own hours. That was the point of writing the number down.
- Where did the review come from? The E-Team on February 1, 2022 asked for cost to serve measured for 3PL plus 4PL against 4PL only, and for the visibility team to publish its own weekly review. By February 14 the shell existed, with arrival accuracy at the discharge port and interaction elimination as the first two lines. The sheet's rules were five weeks of trend and red or green at 5 percent relative variance.

4.5 Flexport — freight-forwarding app, partner route, reuse, eighteen months to six

Stakeholder and trade-off. Say the five million once, as avoided cost, never as an annual reduction. This was not a decision to hire a vendor instead of building. It was which persona went first, and which modules were reused, on our own app.

SITUATION/TASK

At Flexport the freight-forwarding app had an inherited export roadmap: operators first, eighteen months, already shown to leadership, my manager's name on it. Engineering capacity was fixed. The baselines were 339.5 minutes of operator time and

about 2,200 operator interactions on a departed shipment. Forwarding partners would wait. I owned the sequencing recommendation, not the roadmap. I restated the question as time to value and cost per shipment.

ACTION

I priced both sequences at the work-item grain. Operator-first cut cost to serve about 25 percent in nine months and left operators a fifth of their time in the old app. Partner-first cut it about 50 percent in six months, on a chassis the operators would reuse. Anything over eight developer-weeks to rebuild was reused. I read it to my manager privately. He argued two points, I changed one, and engineering had two days to break it. Operators' two further quarters on the old screens were priced in the open, with a reversal trigger, and he put his name on the flip.

RESULTS

First value came in month six instead of month eighteen, at 66 percent less engineering effort. About five million dollars of avoided cost, said once. The import-side app went live in April 2022 with eleven partners in ten countries, and 48 of 53 import partners moved within a month. The export scope was the later launch. I do not call April 2022 the export launch.

MATH

The five million is the record, one avoidance figure, said once. The build-up below is a model. It lands within about twelve percent of the record. It does not replace the record. Twelve engineers and twenty thousand shipments are not a headcount or a volume I will call measured.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Cost per shipment = minutes x blended rate	339.5 x $0.45. The $0.45 is a labeled blend of pool rates $0.56, $0.61, and $0.24 a minute. Record inputs. The blend is derived.	$152.78, spoken as about $153
2	Unit saving, partner-first	153 x 50 percent. The 50 percent is the record re-estimate at the work-item grain.	about $76 a shipment. Operator-first is 153 x 25 percent = about $38, and it is capped by the 20 percent of time still in the old app.
3	Calendar delta	18 - 6	12 months earlier. Record.
4	Effort to first value	100 - 66	34 percent of the inherited path. The 66 percent reduction is record.
5	Engineering avoided, model	12 engineers x 18 months = 216 engineer-months. 216 x 66 percent = 142.56. 142.56 x $20,000. The twelve and the $20,000 are mine. The rate sits in a $15,000 to $25,000 benchmark band.	$2,851,200, about $2.9 million, one time, modeled
6	Run-rate pulled forward, model	$76 x 20,000 export shipments = $1,520,000 a year. The 20,000 is illustrative. Twelve months of that run-rate = $1,520,000.	about $1.5 million pulled forward, modeled. This is not the lane program's $1.5 million and not a pre-production estimate. Say the noun.
7	Model versus the record	2.85 million + 1.52 million = 4.37 million, which the packet rounds to $4.4 million. (5.0 - 4.4) / 5.0	12 percent under the record $5 million. Say the five million once. Call the split a model. Do not also say "annual reduction."
8	Confidence buffer, if asked	(10 x 1.0 + 10 x 1.4) x 1.15. Two illustrative ten-week modules. The 1.0, 1.4, and 15 percent testing buffer are the record rule.	27.6 weeks against 20 naive. Buffers inflate the low-confidence items only.
9	Import partners	48 / 53	90.6 percent, the documents print 90.5 percent. Say 48 of 53 import partners in a month. Not export partners.
10	Scope share of the partner-first slice, record-from-docs	FFA Options Summary, comprehensive scoping: total scoped 905.9 engineer-months, which the sheet converts to 75.5 engineer-years. P0 Export Partners FCL: 126.9 engineer-months, 10.6 engineer-years. 126.9 / 905.9.	14 percent of the whole scoped app. The partner-first launch was one seventh of the scope, on the chassis the rest reused. P1 alone was 219.4 engineer-months.
11	Testing buffer, record-from-docs	The same sheet notes "+15 percent effort in roadmap to account for pre-launch testing." Matches the 1.15 in step 8.	The buffer rule you quote is the sheet's rule, not yours alone.

SOLUTION APPROACH

Inputs were minutes and interactions per departed shipment, engineering re-estimates, and a legacy residue of twenty percent. The model was cost per shipment, then two sequences at the work-item grain. The decision rule was partner-first because it cut unit cost by half in six months rather than by a quarter in nine, plus reuse above eight developer-weeks. I simplified by refusing to treat the roadmap as mine and by swapping scope instead of absorbing it. I extended by a written reversal trigger and by pricing

the operator delay in the open. The long-term risk was a flip that died when I left the room. His name on it, and a burn-up from 55.8 percent to 83.2 percent with every work item owned, are what kept it.

IF THEY PROBE

- Was this build versus buy? No. Partners were the first persona. The reuse rule decided which modules were rebuilt and which were reused from our own app. The separate build-versus-buy call was on visibility, where we did not buy a competitor's data.
- What would you do differently? I would not change the private-first order. I would have put the operator quarters on the first page, not the second, so the objection was answered before it was asked.
- What did it cost? Operators stayed on old screens for two more quarters. That cost was priced and published. It was not free. The five million is avoidance, engineering not spent plus value pulled forward, said once.
- How did your manager react? He asked for the estimation, argued two points, and put his name on the flip. That is why it was private first.
- Is the five million annual? No. The modeled run-rate under it is about $1.5 million a year on an illustrative shipment count. I never present the avoidance as that run-rate.

4.6 Flexport — carrier bid, six weeks to about three

Short. Ownership. The cycle time is a user anchor. There is no award dollar. Use this only if he asks for a process you shortened, or for a time you refused a tool.

SITUATION/TASK

A transportation team ran a carrier bid twice a year, hundreds of lanes and dozens of rate files, about six weeks a cycle. Everyone wanted a better optimizer. I owned the requirements, the constraint rules, and a parallel run. I did not own the award.

ACTION

I sat through one full cycle and logged where the hours went. Most of the six weeks was reconciliation, because every carrier named lanes and modes differently. I specified a lane master on origin, destination, and mode, wrote each constraint as a rule a person could read, and set a gate: the tool had to match the last two real awards within five percent before its output counted. One region parallel-ran a live bid. A rate file older than ten days carried a stale banner.

RESULTS

The cycle went from about six weeks to about three, and the pilot region finished about three weeks early. Other regions picked it up without a mandate. I claim no award savings. Procurement joined late, and two of their rules had to be added before the second region. That is the miss.

MATH

STEP	FORMULA	SUBSTITUTION	RESULT
1	Cycle delta	about 6 weeks to about 3 weeks. Both ends are a user anchor, from my time audit, not an audited report.	about 3 weeks out, about half the cycle
2	Share removed	(6 - 3) / 6	0.50 on those anchors. Say "about half." Do not pretend the hours were instrumented by activity.
3	Trust gate	mismatch against the last two awards	under 5 percent, or the planners kept the spreadsheet. User anchor.
4	What is not on record	An award dollar. A split of the six weeks into hours by activity. A loaded-rate saving.	Do not lead with a dollar. The award belonged to transportation and procurement.

SOLUTION APPROACH

The input that mattered was the time audit, not the rate file. The model was a lane key plus explicit constraints, not a new solver. The decision rule was a five percent back-test before anyone trusted the output. I simplified by refusing the optimizer until the names matched. I did not extend the tool into the award. The long-term risk was a stale rate file silently winning a lane. The ten-day banner is that control. I would bring procurement into the audit, not the design review.

IF THEY PROBE

- What would you do differently? Procurement in the room at the time audit. They owned two rules I had to retrofit.
- What did it cost? One full cycle of watching before any build, and some impatience from a team that wanted a tool. No award dollar.

- How do you know it was not a quieter cycle? Same team, same kind of bid, against my own audit of the previous cycle. I say "about half" because those counts are my anchors.
- Why five percent? Below that, planners stopped re-keying. Above it, they would have kept the old file open beside the tool.

4.7 Elementum — Austria, data quality to about 99 percent

Short. Ownership. Say "within a quarter." The master packet says about three months. The Apple workbook says inside two months. "Within a quarter" is true under both. Do not say four hundred plus lanes. The five-hundred-thousand and the eight million are Finance-signed, executed shipments only.

SITUATION/TASK

At the customer's Austria site, transportation could not measure on-time delivery. The report said data quality was 40 percent. The team spent eight to ten hours a week reconciling dates by hand. The site, the carrier teams, and my sponsor all believed the carriers were late. I owned the analysis and the rules. The data team built the engine.

ACTION

I pulled three years of shipments in SQL and ran a defect census: missing rates, lanes at the wrong grain, five spellings of one carrier, brochure transit times posing as promises. True quality was about 55 percent, not 40, and the defect was measurement. I specified the promised date as a routing-guide lookup on ship-from, ship-to, mode, volume band, Incoterm, and service, and a miss paged a person instead of defaulting a carrier. Finance signed the savings method before any dollar: contracted guide rate minus paid rate, executed shipments only. I would not release the carrier scorecard until the automated on-time figure matched a manual sample.

RESULTS

Data quality went from about 55 percent true to about 99 percent within a quarter. The carriers did not dispute the scorecard. Austria was about $500,000 a year on that signed method. The same rules file, plus one data-quality cycle a site, then rolled to twenty-five European locations, about $8 million, executed only. Manual reconciliation fell from eight to ten hours a week. The packet also prints one to two hours after. I say "a small fraction" unless I am ready to label the one-to-two as the figure my earlier record carries.

MATH

STEP	FORMULA	SUBSTITUTION	RESULT
1	The reframe	Reported quality 40 percent. True quality at the measured grain about 55 percent. Both record.	The 40 percent was a single-field proxy. The problem was not the carriers.
2	After	about 99 percent	Within a quarter. Do not pick two months or three months unless he forces the question. If he does, say the sources disagree and the quarter is the line you will stand on.
3	Austria dollars	$500,000 a year. The packet ties it to 10 percent of a roughly $5 million freight base. The base is a user anchor. 500,000 / 5,000,000 = 0.10.	Say five hundred thousand a year at Austria. The base is the anchor, not a second result.
4	Network dollars	about $8 million across 25 European locations, executed shipments only. 8,000,000 / 25 = 320,000.	$320,000 a site is a derived average, not a measured site result. Do not say it as if each site was audited at that number. Do not say a lane count for this program.
5	Hours	8 to 10 hours a week before, record. After: the packet prints 1 to 2 hours. Reconciliation marks that after-state as a user anchor, because the other source only says the hours were gone.	Say eight to ten hours down to a small fraction, unless you are explicitly labeling the one-to-two.

SOLUTION APPROACH

The input was three years of history, not the carrier's explanation. The model was a promised date computed from the routing guide, with Incoterm in the key on the lanes where it changes who owns the clock. The decision rule was no scorecard until it matched a manual sample, and no dollar until Finance had signed the method. I simplified by refusing a carrier-performance product. The defect was our lane and promise data. I extended one site's rules file to twenty-five sites with a one-page rollout checklist, not a new design per site. The long-term risk was a scorecard the carriers would win by disputing the measure. Holding the scorecard until the sample matched is what retired that risk.

IF THEY PROBE

- What would you do differently? Start from an industry transit benchmark only as a hypothesis. I proved the brochure wrong lane by lane, which was slower than it needed to be. The census was still the right first week.
- What did it cost? A sponsor who was unhappy for weeks, and the manual reconciliation kept alive until the gate passed. Negotiated-but-not-executed freight never counted, so the dollar is smaller than a slide might want.
- Why not ship the scorecard on the data you had? A scorecard on 55 percent data would have been disputed, and the carriers would have been right.
- What did Finance sign? Contracted guide rate minus paid rate, executed shipments only, before the number existed.

4.8 Back-up — Sephora S/4HANA, one published available-for-sale

Back-up only. Stakeholder. Use if he asks what you do now, or how you get a definition signed before anyone configures it. One umbrella program with 4.9 if both come up.

SITUATION/TASK

Since April 2023 I have led Sephora's inventory data contract. Available had three definitions: merchandising wanted its reserve promisable, finance wanted on-hand to tie to the book, and dotcom wanted in-transit promisable. Engineering teams each wanted to compute the number. I owned no system. The target was one signed available-for-sale per node and channel, and oversell at zero.

ACTION

The reserve stayed visible and not promisable. The promising basis is the lower of the two books. My first in-transit rule, promise none of it, starved dotcom, so I replaced it with a lane window, an on-time floor, and a kill switch. The order system reads the published number and never computes its own.

RESULTS

Three organizations signed on June 5, 2025. Oversell went from recurring to zero through both go-lives. No SKU carried two published values. The correction did not raise oversell, and the kill switch did not fire.

MATH

STEP	FORMULA	SUBSTITUTION	RESULT
1	Oversell	Recurring before. No incident count is on record. After: zero through two go-lives. Record.	Say recurring to zero. Do not invent a baseline count.
2	In-transit window, promise math	lane sigma about 0.3 day, user anchor. Window about 1.645 x 0.3.	0.4935 day, spoken as about half a day, about 95 percent coverage. Not a measured sigma I will call audited.
3	What the wrong rule cost	No dollar on record for the conversion lost.	Not on record. Do not invent one. The movement I will claim is starved lanes restored, oversell increase zero.
4	Who computes	One publisher. The order system reads it.	Consumers never compute. That is the design, record.

SOLUTION APPROACH

Inputs were the stock buckets, who owned each, and which customer is hurt if a bucket is promised. The model was one published number, reconciled daily to the book at tolerance zero, not merged into it. The decision rule was physical truth in the ledger, promisable truth in the published number. I simplified by refusing a design where every consumer computes. I extended the in-transit rule only after the aging report showed the blanket ban was wrong. The long-term risk is a side configuration. The same three signatures had to sign the correction.

IF THEY PROBE

- What would you do differently? Use the lane window on day one. A blanket ban is a claim that every lane has infinite variability, and almost none do.
- What did it cost? No dollar on record for the lost conversion. I will say that. The correction did not raise oversell.
- Why not let the ledger be the promising truth? It posts documents and settles store sales as a nightly aggregate. Promising needs every click and scan. The two are reconciled, not merged.
- Did you escalate? No. Private first, then the three signatures.

4.9 Back-up — Sephora, dual inventory, council to hypercare

Back-up only. Cross-functional, complex timelines. Same umbrella program as 4.8. Say so if both are in the room.

SITUATION/TASK

Sephora's Utah building ran as two warehouses in one building. One pool shared a path with a system upgrade and a robotics ramp, under an October-to-December peak freeze, about thirty tracks and sixty-plus stakeholders. I owned the gates. Zero customer-facing incidents was the bar.

ACTION

Discovery ran March 29 to April 30. The council got three decisions and a fence: challenge May 15, signed effort July 2, sign-off July 31, specification August 14. I split the release: inbound in January 2025, inventory in May. A three-week robotics slip went to steering that week. Week two missed a 98 percent pick gate, so I held the robots to about 30 percent of volume until they cleared it.

RESULTS

Both go-lives had zero customer-facing incidents. Processing efficiency finished up 15 percent, the operation's own number. The log closed 714 of 836 rows. After May it spiked to 271 and burned down 65, 21, 14, 4. Full automation was back in week three.

MATH

STEP	FORMULA	SUBSTITUTION	RESULT
1	Close rate	714 / 836	85.41 percent, spoken as 85.4 percent. Record.
2	The 15 percent	Processing efficiency, the operation's signed number.	Not on record: the before and after rates that produce the 15 points. Say their number. Do not reconstruct it.
3	Incidents	Zero customer-facing through both go-lives.	Not on record: a baseline incident count. Do not invent one.
4	Gate versus week two	98 percent bar, 99 percent stretch. Week two printed 94 percent. About 30 percent of volume stayed on carts.	Full automation in week three, after the gate cleared. Record.

SOLUTION APPROACH

Inputs were the requirement rows, the peak freeze, and a vendor date that was not the floor's date. The model was two go-lives, each with one defect class, and one log as the gate. The decision rule was business criticality times effort, and a robotics share that drops when accuracy misses the bar. I simplified by fencing reserve-math changes out of the release. I extended the test rather than the courage. The long-term risk was two live truths during the interim. Each had an owner and a trigger before the council signed.

IF THEY PROBE

- What would you do differently? A readiness row for the robotics date, owned in November, not discovered in December. I handled the slip in the open. I saw it late.
- What did it cost? Analysts lived with two books for the interim, about four months. The reserve improvements waited, priced as a later enhancement. The record carries no program dollar I will lead with.
- Same program as the inventory contract? Yes. Same two go-lives, same log, same gate grammar. Say one umbrella program.
- How did you know hypercare was over? The spike decayed 271, 65, 21, 14, 4, and blockers stayed under the bar. A calendar date does not close a defect.

4.10 Failure — the security review I under-planned

Flexport, inside the visibility pipeline. Use this when he asks for a failure. Own it in the first sentence. Say what changed. Stop.

SITUATION/TASK

On the Shipment Visibility pipeline, ticket data carries customer content. I treated the access review as an engineering concern. It is a program dependency with a lead time. I found that out mid-build, not at kickoff.

ACTION

The day it surfaced I told leadership the gap was mine, not a delay by the security team. I re-sequenced the work that did not touch customer content, kept the Monday review on the manual path, and did not skip the review to make the date. The parallel run absorbed the two weeks. Cutover still waited on both gates: access, and parity within one percent.

RESULTS

No Monday was skipped. The pipeline still reached the 60 to 90 percent accuracy and the one percent parity gate in story 4.4. The permanent change is a kickoff question: whose approval does this data need, and what is its lead time. I plan that lead time. I do not discover it.

MATH

STEP	FORMULA	SUBSTITUTION	RESULT
1	Schedule miss	Access review under-planned by two weeks. Record.	Two weeks on the manual path. The parallel run covered it.
2	What that two weeks is not	No dollar for the two weeks is on record. The $103,680 in story 4.4 prices the old weekly process for a year. It is not the cost of this miss.	Not on record: a dollar cost of the two weeks. Do not invent one. Say two weeks, owned the same day, no Monday skipped.

SOLUTION APPROACH

I had scoped correctness and omitted approval. The decision rule now is that a data feed does not enter a plan until the access lead time is on the same page as the build. I simplified the recovery by keeping the manual path, which already worked, instead of inventing a temporary feed. The long-term risk was a team that skips the review to save a date. Writing the question into every later kickoff is the control.

IF THEY PROBE

- What would you do differently? The review is a line on the kickoff, with a named owner and a lead time, before the first ticket is modeled.
- What did it cost? Two weeks of schedule. No skipped Monday. No dollar I will invent.
- Who did you tell? Leadership, the same day, in the first line: the miss was mine.
- Did the customer see it? The Monday decision kept running on the manual numbers. The cutover did not move until both gates were green.

4.11 Design cases — not delivered results

Say "design case" in the first sentence. No production metric. The Walmart material is a hiring-prep workbook of illustrative numbers, not a client result. The Google Fiber material is a designed infrastructure flow, not a network you ran.

SITUATION/TASK

Two design cases, labeled as such. A marketplace promise is handling plus transit plus a buffer, and a flat buffer mis-prices both the tight lanes and the volatile ones. An infrastructure build treats a requested date as if it were confirmed, across the purchase order, the warehouse, the van, and the return.

ACTION

I would set the buffer off the measured spread, one signed dial, shadowed, and promoted only if promise-kept still holds. Peak resets the dial before peak. On the build network I would run one document per hop, and a vendor would not activate until the message is valid and the transaction succeeds. An acknowledgment with no date is not supply.

RESULTS

No delivered result. The instrument is one dial, one gate, one owner, one rollback. If he asks what I have actually done, I switch to the two hundred measured lanes.

MATH

> Every figure in this table is a design-case model or a design bar. None of it shipped. Do not say these as results. If he does not ask you to walk the model, do not open it.

STEP	FORMULA	SUBSTITUTION	RESULT
1	Delivered metric	None. Walmart pre-purchase and Google Fiber are design cases in the packet.	Not on record: a production accuracy, a production dollar, a production lead time.
2	Promise, if asked to walk the model	buffer = k x sigma. The illustrative dial uses k = 1.645, the 95 percent point of a normal curve. The packet's twelve-cohort model is illustrative.	Stay in the model. Do not say a fast-share point or a weekly dollar as something you delivered.

STEP	FORMULA	SUBSTITUTION	RESULT
3	Infrastructure lead time, if asked	The design writes node equipment at 12 to 40 weeks. Illustrative design input.	A bar I would set, not a result: valid advance ship notices at or above 95 percent before a vendor activates, which is the gate I have actually run on carrier onboarding. Unconfirmed after 48 hours stays out of trusted capacity. Any crew-idle dollar on that page is illustrative. I would ask Finance before I sized it.

```
design case only — not a flow I operated

vendor signal -> dated confirmation -> serialized ASN -> receipt
      -> field asset -> return disposition -> settlement
miss on any hop: quarantine, do not default a date
capacity = confirmed supply, never the requested date
```

SOLUTION APPROACH

The promise problem decomposes into handling, transit, and buffer, and I would not spend a buffer day to hide a sourcing choice or a bad inventory feed. The infrastructure problem decomposes into a document per hop and a gate per node type. In both, I simplify to one dial or one date chain, and I extend with a shadow period and a kill switch rather than a big-bang config. The long-term risk is a green average that hides a bad cohort. Cohort gates, not a portfolio average, are the mitigation. That is the same habit as holding a wave and holding a scorecard.

IF THEY PROBE

- Have you done this? The design, yes. The production result, no. The lived version is measured lane lead time, and a promise rule I set too tight and then corrected.
- What would you do differently in the design? Shadow against real orders, and require both "the message is valid" and "the business transaction succeeded" before a vendor or a carrier graduates. I once measured completeness from day one and tender acceptance only from a later cohort, so a carrier-data issue showed up late.
- What did the design cost? Nothing delivered, so nothing saved. The price of the approach is a slower first month for a weak data source, on purpose.
- How does this meet a critical pull time? The pull time is the confirmed clock, not the requested one. I would rather publish a later promise that the network can make than a faster one the sort will miss.

5. Trade-offs I would expect him to ask about

He has a Lean Six Sigma and operations-research background, and he has said he wants the limitation of a product early. Lead with the rule, then the story, then what you would check here. Do not lecture the Amazon network.

TRADE-OFF	HOW I FRAMED IT IN A PAST STORY	THE DECISION RULE I USED	WHAT I WOULD CHECK AT AMAZON
Speed versus cost	4.1. A faster brochure promise versus a tail I could defend. Most lanes released stock. The fat-tail lanes took stock in.	Promise at the measured ninety-ninth percentile. Release cost only where the measured spread sits inside the old assumption. Alert earlier, at the ninetieth, so a re-plan is still cheaper than an expedite.	Whether a critical pull time is set from the lane's actual distribution or from the service we wish we had, and which building pays when the pull time is missed.
Consolidation versus frequency	4.1 and 4.3. A thousand thin lanes, or the lanes that carry the spend, fully measured. At Flexport, all customers now, or a wave when the gates hold.	Spend share first, same gate for the remainder. A wave boards only at the current load. Volume alone does not earn the next wave.	Whether a network change is approved on lane count or on the share of packages and the share of cost it actually moves.
Central versus regional stock	4.1 and 4.7. One pad on every lane, or a buffer per lane. One national carrier story, or a rules file proven at one site and then copied.	The buffer names what it is protecting, demand or transit, at that lane. A site does not go live on a default carrier.	The public 2023 shift to eight US regions is the version of this I have read. I would ask how much inventory the regions had to hold so the middle mile could get shorter, and who signs that trade.
Build versus partner	4.5. Not a vendor decision. Operators first and build the path, or partners first and reuse the chassis. On visibility, we did not buy a competitor's data.	Whichever sequence cuts cost per shipment sooner, with reuse above eight developer-weeks, and the operator delay priced in the open.	Where TITANS should build config automation, and where the team should keep a person's gate because the limitation of the product is already known.
Automate versus manual	4.3 and 4.4. A ticket stayed in the loop until the source held. The Monday review stayed manual until parity held.	A source leaves the human loop at or above 95 percent for four weeks and re-enters below 90. Cut over only within 1 percent for consecutive weeks.	Which configs a model may write, which a person must still sign, and what the kill switch is when the model is wrong in a way the team has already seen.
Standard versus custom config	4.1 and 4.7. One advance-ship-notice standard, with a portal floor for a small carrier. One routing-guide rules file, and a miss pages a person instead of a custom default.	The standard is the data, not the transport. No silent default. A custom map is a hold, not a favor.	How a one-off sort pattern or a one-off pickup gets into production, and whether it expires or lives forever as an exception the next change has to route around.
Keep or deprecate, when two scoreboards disagree	4.3b. Crux at 63 against the 4PL platform at 25, on different containers and different horizons. Normalize first to earn the time, then improve to deserve it.	Reconcile before compare: same containers, same sources, same week, same horizon. Name the residual against yourself in the same readout. One forum signs the improvement plan and the retirement plan together.	When two systems report the same lane or the same pull time differently, which one the network is configured from, who owns the reconciliation, and whether a model's recommendation and a planner's override are scored on the same basis before one of them is retired.

Part 3 — Amazon transportation, end to end

I. Amazon transportation, end to end

This section is the physical path a unit takes, inbound and outbound, and the decisions that sit on that path. Amazon has published the shape of the network: fulfillment centers, sort centers, delivery stations, air hubs, a first-party middle mile, and a last mile run largely by its own logistics. It has not published the clock times, trailer rates, or cost-per-package stack used below. Those worked examples are the generic model, with every assumption stated. Where a figure is Amazon's, the year and the source are in the sentence and again in the list at the end.

I.1 Inbound: vendor and seller freight to a promiseable unit

Two inbound customers share the buildings. A first-party vendor sells the unit to Amazon. A third-party seller keeps title and uses Fulfillment by Amazon, so Amazon stores and ships it. In both cases the transportation question is the same: which building receives the freight, and who pays to put the units near the customers who will buy them.

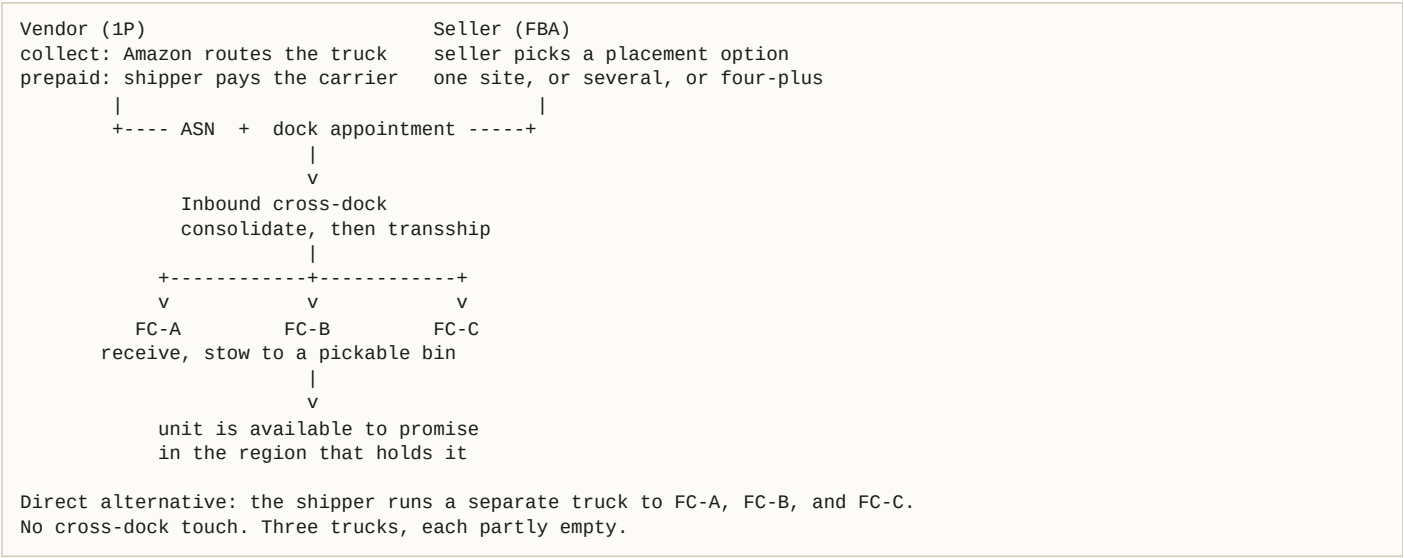
Amazon's Carrier Central manual treats an inbound purchase order as collect or prepaid, and it will not book the two on the same shipment line. An Amazon reference number is mandatory for collect freight; the manual says that number is assigned to collect vendors, and that the purchase order has to show as collect in Amazon's system before an appointment is granted. Prepaid purchase orders go on their own carrier reference. In ordinary freight language, collect means the receiver controls and pays the carrier, and prepaid means the shipper does. The manual confirms the two types and the appointment rule. It does not publish a rate card.

The same manual is the public description of dock discipline. The carrier requests an appointment with the destination, the purchase orders, the pallet and carton counts, and a bill-of-lading or seller reference so Amazon can tie the truck to the advance ship notice. Amazon grants a time. A purchase order pointed at the wrong building, a collect order mixed onto a prepaid line, or a reference that does not match the advance ship notice is rejected or becomes a defect. The generic operating point, which the manual supports without using these words, is that the yard plan and the receive labor are built from the appointment, and the inventory record is built from the advance ship notice. A truck that arrives outside that pair consumes a door and does not become promiseable inventory on time.

Where the truck goes is the placement decision. A 2016 MIT thesis on Amazon transshipment describes the domestic cross-dock: the vendor sends the network quantity to the cross-dock, and the cross-dock splits it and transships to the destination fulfillment centers, instead of the vendor running a separate truck to each one. A 2019 FHWA briefing describes the same kind of building, the inbound cross-dock, close to ports and rail, destuffing containers and loading fulfillment-center trucks. This packet calls that node the inbound cross-dock. A distinct building officially named a receive center was not verified, so that name is not used as an Amazon fact.

The economic signal for sellers is public. In its late-2023 fee announcement, Amazon said it would charge an inbound placement service fee, effective March 1, 2024, averaging $0.27 per unit for standard-size products and $1.58 per unit for Large Bulky products, because placing inventory close to customers improves speed and lowers transportation cost. The seller can pay a reduced fee or no fee by shipping to more locations. Amazon's seller-facing explanation of the options is: no fee when the seller sends inventory to the recommended set of locations, generally four or more; a reduced fee for a partial split, generally two or three; and the fee when the seller sends everything to one location and Amazon spreads it. Outbound fulfillment fees were cut at the same time, by an average of $0.20 per unit for standard-size products and $0.61 for Large Bulky, starting April 15, 2024. Read together, Amazon unbundled the cost of placement from the cost of customer delivery, and handed the seller the choice.

Jassy's 2023 shareholder letter, published in 2024, named the same lever from the company side: inbound fulfillment architecture and the resulting inventory placement were a focus for 2024, after regionalization had already cut transportation distance. Placement is how a unit becomes available against the promise in the region where the order will be placed. Receive and stow are the building steps that make that true in the system. The generic model is: until the unit is received and stowed to a pickable location, it cannot be promised, however close the truck got.



The dollars below are a teaching model of that picture. They are not Amazon's rates, and they are not the inbound placement fee. The fee is a published price. This stack is an illustration of why a price like that exists.

WORKED EXAMPLE: THREE PARTIAL TRUCKS, OR ONE INBOUND TRUCK AND A SHARED ONWARD TRAILER

Assumptions (illustrative unless cited): 900 cases; a trailer holds 900 cases; linehaul costs $1.50 per truck-mile and nothing else; the three fulfillment centers are 300, 900, and 1,500 miles from the vendor; the inbound cross-dock is 60 miles from the vendor; a direct ship is one truck per center at 300/900 fill; through the cross-dock, the 900 cases fill the inbound truck and then ride as 300/900 of a full trailer on each onward lane; the cross-dock touch is $0.40 per case.

STEP	FORMULA	SUBSTITUTION	RESULT
1. Direct truck count	one truck per destination	3 centers	3 trucks
2. Direct miles	d1 + d2 + d3	300 + 900 + 1,500	2,700 truck-miles
3. Direct fill	cases per truck / trailer	300 / 900	33 percent
4. Direct cost	miles x rate	2,700 x $1.50	$4,050
5. Direct cost per case	cost / cases	4,050 / 900	$4.50
6. Cross-dock inbound	miles x rate	60 x $1.50	$90
7. Onward share of each trailer	cases on the lane / trailer	300 / 900	1/3 of each onward truck
8. Onward miles attributed	(d1 + d2 + d3) x share	2,700 x 1/3	900 truck-miles
9. Onward cost	attributed miles x rate	900 x $1.50	$1,350
10. Transport total	inbound + onward	90 + 1,350	$1,440
11. Transport per case	transport / cases	1,440 / 900	$1.60
12. Cross-dock touch	fee per case x cases	$0.40 x 900	$360
13. All-in through the cross-dock	transport + touch	1,440 + 360	$1,800
14. All-in per case	all-in / cases	1,800 / 900	$2.00
15. Truck-equivalents	inbound trucks + onward share	1 + 3 x (1/3)	2 truck-equivalents
16. Savings per case	direct - cross-dock	$4.50 - $2.00	$2.50 per case

The direct path uses three trucks and 2,700 miles because each lane pays for a whole trailer to move a third of one. The cross-dock path uses two truck-equivalents and puts the long miles on trailers that other freight fills. The $2.50 gap is the room in which a placement fee, a vendor allowance, or an Amazon-paid transship can sit. The 2024 fee is the published version of that choice for sellers: pay Amazon to do the spread, or spend your own freight to deliver into more buildings and pay less fee, down to zero.

SAY THIS — 90 WORDS

Inbound is a placement decision before it is a freight decision. A vendor or a seller can pay to deliver cases into the fulfillment centers that will ship them, or pay Amazon, through the inbound placement fee, to spread one receipt across the network. The fee is the price of that spread. I would check three things before a shipment moves: the advance ship notice matches the appointment, the destination matches where the demand is, and the cost per case of a partial truck is visible next to the fee.

I.2 Outbound: promise, pull time, sort, last mile

A customer order becomes a promise: a ship option and a delivery date. Amazon's customer help page says the date is the shipping date plus transit time, in business days, with an "order within" countdown when the clock is still open. That promise is the constraint. Everything after it is a path chosen so the attempt happens on or before that date, at a cost the network can live with.

Amazon Science (October 14, 2022) describes the path in the first-party network. Packages leave fulfillment centers for sort centers, which aggregate shipments from several fulfillment centers, and then go to delivery stations, where vans are loaded for the last mile. The connections are lanes, air and ground. The same article names the label step: SLAM, scan, label, apply, manifest. Amazon's own fulfillment-center tour uses the same expansion and places it after pack and before the outbound trailer: the package is identified, weighed, labeled, and manifested, then sorted to a trailer. In the generic model the label is the moment the package is committed to a lane and a departure. Change the lane after that and you are rehandling.

The sort center, in Amazon's public description, sorts by destination and hands trailers to delivery stations or to a carrier partner, including the local post office. A 2025 paper by Amazon authors in INFORMS Journal on Applied Analytics describes a further split that regionalization made useful: hubs, which protect trailer fill on the long cross-region haul, and last-mile sort centers, which consolidate for delivery stations. Direct fulfillment-center-to-delivery-station trucks are the bypass. Amazon Science (July 24, 2023) says regionalization created more of those direct trucks, because more of a destination's volume now comes from fulfillment centers inside the region, so a trailer can fill without a stop at the sort center.

The last mile is a mix. The 2023 sustainability report says more than two-thirds of Amazon packages that year were delivered by Amazon's own logistics network. The public programs on that network are Delivery Service Partners, independent businesses that run the routes; Amazon Flex, drivers using their own vehicles who pick up at a station and deliver a block, typically three to six hours for Amazon.com packages; and, on the rural-network page, Hub Delivery with local businesses. Amazon Science (2022) is explicit about the rest of the history: before the first-party network, deliveries were completed by partners such as the U.S. Postal Service and UPS. The 2025 Form 10-K still says Amazon relies on a limited number of shipping companies to deliver completed orders. A current split among DSP, Flex, USPS, and UPS was not published, so it is not stated here.

Same-day sites shorten this path by collapsing it. Doug Herrington, July 31, 2023: same-day buildings are smaller, sit near the metros they serve, pick directly to pack, and fulfill, sort, and deliver from one site. He gave the average time from pick to a package on the outbound dock as 11 minutes, more than an hour faster than a traditional fulfillment center, and said same-day was then available across more than 90 U.S. metro areas. Jassy's 2023 letter, published in 2024, counted 58 of these buildings, holding the top 100,000 SKUs, with millions of other SKUs injectable from nearby fulfillment centers, and called them the lowest cost to serve in the network. The 2025 letter calls them Same Day Fulfillment Centers, or SSDs: more than 85 in the U.S., carrying the top 90,000 SKUs. The phrase "sub-same-day" was not verified as Amazon's official expansion of SSD. The operating fact that was verified is the collapsed path: one building, no separate sort center, a very short clock.

Air is the long-lane version of the same idea. Amazon's August 11, 2021 press release opened the Amazon Air hub at Cincinnati/Northern Kentucky International Airport as the primary U.S. hub, an investment of $1.5 billion, with an 800,000-square-foot sort building meant to process millions of packages a week. The 2023 sustainability report says in-region shipping reduced reliance on air. Air remains the tool for distance that ground cannot cover inside the promise.

Amazon Freight and Relay are the middle-mile commercial layer, not the last mile. Amazon's supply-chain site distinguishes them: Amazon Freight, in the Relay for Shippers portal, is an inbound less-than-truckload and truckload option, including to non-Amazon destinations in the lower 48; Relay is the capacity side, the trailers and the carriers that haul them. The aboutamazon transportation article describes Relay as the tool small carriers use to book Amazon loads. Treat Freight as a way freight gets into and across the network, and Relay as how truck capacity is matched to those loads.

```
Order
  -> promise: ship option and delivery date
  -> assign a fulfillment center that can hit the date
  -> pick, pack
  -> SLAM: scan, label, apply, manifest
  -> outbound dock
  -> CPT: last departure that still makes the promise
  -> linehaul to a sort center (or direct to a delivery station)
  -> sort by destination
  -> linehaul to the delivery station
  -> sort onto a route
  -> last mile: DSP van, Flex block, or a carrier (USPS, UPS)
  -> delivery attempt on or before the promised date

Same-day building: pick -> pack -> sort -> dispatch, one site.
The sort-center leg is the one this path deletes.
```

Critical pull time is the outbound clock. Amazon does not define it on a public retail page. A 2016 MIT thesis on Amazon operations defines CPT as the time an item must be picked, palletized, and on a truck, and defines order critical pull time as the time the package must depart Amazon, set from the ship-method cycle time. That is the definition used here: the last time a package can leave the fulfillment center and still make the promised delivery. The hours below are the generic model. They were not taken from an Amazon standard.

The cascade runs backward from the delivery window. The van has to leave the delivery station early enough to finish the route inside the window. The package has to be through the station's inbound sort before that departure. It therefore has to have left the sort center one linehaul earlier, and it has to have arrived at the sort center before that building's sort window. It therefore has to have left the fulfillment center one linehaul before that. The pack-to-dock buffer sits in front of the departure: a package still at pack when the trailer closes has missed the pull time even if the trailer itself is on time. Miss the pull time and the package waits for the next departure, which is the next promise, unless a recovery truck or air is paid for.

```
20:00 Tue   promised delivery, last stop of the route
09:00 Tue   van leaves the delivery station
07:00 Tue   package is at the delivery station        minus 2 h station sort
03:00 Tue   package leaves the sort center            minus 4 h linehaul
00:00 Tue   package arrives at the sort center        minus 3 h sort window
18:00 Mon   CPT: package leaves the fulfillment center minus 6 h linehaul
16:00 Mon   pack is complete                          minus 2 h dock buffer
```

WORKED EXAMPLE: BACKWARD SCHEDULE TO THE CRITICAL PULL TIME

Assumptions (illustrative unless cited): delivery is promised by 8:00 p.m. Tuesday; the route that can meet that promise starts at 9:00 a.m. Tuesday, so the schedule runs back from 9:00 a.m. and not from 8:00 p.m.; delivery-station inbound sort is 2 hours; sort-center-to-station linehaul is 4 hours; sort-center sort window is 3 hours; fulfillment-center-to-sort-center linehaul is 6 hours; pack-to-dock buffer is 2 hours.

STEP	FORMULA	SUBSTITUTION	RESULT
1. Anchor	route start, not the end of the window	promise 8:00 p.m. Tuesday, van leaves 9:00 a.m. Tuesday	9:00 a.m. Tuesday
2. Arrive at the delivery station	route start - station sort	9:00 a.m. - 2 h	7:00 a.m. Tuesday
3. Leave the sort center	station arrival - linehaul	7:00 a.m. - 4 h	3:00 a.m. Tuesday
4. Arrive at the sort center	sort-center departure - sort window	3:00 a.m. - 3 h	12:00 a.m. Tuesday
5. CPT, leave the fulfillment center	sort-center arrival - linehaul	12:00 a.m. - 6 h	6:00 p.m. Monday
6. Pack complete	CPT - dock buffer	6:00 p.m. - 2 h	4:00 p.m. Monday
7. Check the chain	sum of the four path hours	2 + 4 + 3 + 6 = 15 h; 9:00 a.m. Tuesday - 15 h	6:00 p.m. Monday

The critical pull time is 6:00 p.m. Monday. Pack has to be finished by 4:00 p.m. Monday or the package misses a trailer that is itself on time. Each upstream hour is inherited. If the sort-center-to-station linehaul runs five hours instead of four, the pull time moves an hour earlier, to 5:00 p.m. Monday, and so does the pack deadline. That is why a configuration change and a transit change are the same kind of event: both move the clock.

Cost per package is the same path in dollars. Amazon has not published this stack. It has published the total: in the 2023 shareholder letter, U.S. cost to serve was down by more than $0.45 per unit versus the prior year, the first global per-unit reduction since 2018. The illustration splits a generic package into fulfillment-center outbound handling, two linehauls, a sort, and the last mile, then shows the two denominators that move the variable pieces: packages on the truck, and packages on the route.

WORKED EXAMPLE: COST PER PACKAGE, AND WHAT FILL AND ROUTE SIZE DO TO IT

Assumptions (illustrative unless cited): fulfillment-center outbound handling is $0.75 per package and sort is $0.30 per package, both unchanged by fill; the fulfillment-center-to-sort-center truck costs $630 and has 1,000 package positions; the sort-center-to-delivery-station truck costs $378 and has 1,000 positions; a last-mile route costs $198; the base case is 70 percent linehaul fill and 150 packages per route; the tighter case is 90 percent fill and 220 packages per route.

STEP	FORMULA	SUBSTITUTION	RESULT
1. Handling	given per package	$0.75	$0.75
2. Sort	given per package	$0.30	$0.30
3. Packages on a truck at 70 percent	fill x positions	0.70 x 1,000	700 packages
4. Linehaul, fulfillment center to sort center, 70 percent	truck cost / packages	$630 / 700	$0.90
5. Linehaul, sort center to delivery station, 70 percent	truck cost / packages	$378 / 700	$0.54
6. Last mile at 150 packages	route cost / packages	$198 / 150	$1.32
7. Base cost per package	sum of the five pieces	0.75 + 0.90 + 0.30 + 0.54 + 1.32	$3.81
8. Packages on a truck at 90 percent	fill x positions	0.90 x 1,000	900 packages
9. Linehaul, fulfillment center to sort center, 90 percent	truck cost / packages	$630 / 900	$0.70
10. Linehaul, sort center to delivery station, 90 percent	truck cost / packages	$378 / 900	$0.42
11. Last mile at 220 packages	route cost / packages	$198 / 220	$0.90
12. Tighter cost per package	sum of the five pieces	0.75 + 0.70 + 0.30 + 0.42 + 0.90	$3.07
13. Difference	base - tighter	$3.81 - $3.07	$0.74 per package

Handling and the sort did not move. The $0.74 came from the denominators: 200 more packages on each truck, and 70 more packages on the route. In the base case the last mile alone is $1.32 of the $3.81, about a third. That is why a delivery-station catchment and a route size belong in a transportation cost review, not only in a last-mile review. The 2023 sustainability report names the real-world version of step 4 and step 9: Amazon cited better truck fill as one of the operating changes that cut emissions per package that year.

SAY THIS — 90 WORDS

Outbound is a promise walked backward onto a clock. The critical pull time is the last moment a package can leave the fulfillment center and make the delivery window, after you subtract the station sort, both linehauls, and the sort-center window. Cost per package is that same path in dollars: handling, two linehauls, a sort, and the last mile. Fill and packages per route are the two numbers that move the linehaul and the last mile. I would put the clock and the cost on one page before changing either.

I.3 Where a transportation-engineering team sits

Three kinds of decision sit on this network. Configuration is the standing rule a package is executed against: which sort code maps to which building, which ship method is offered, what the critical pull time is, which lane a trailer takes, which ZIP codes belong to a delivery station, and what the customer cutoff is. Planning is the volume that rule will see: the forecast, the labor, and the truck capacity. Design is a change to the structure the rules run on: a new node type, a new building, or a new region boundary. Amazon's 2025 INFORMS paper is explicit that regionalization required all three at once: software changes to inventory, promising, and order assignment; changes to fulfillment-center configuration, network connectivity, and transportation schedules; and a new partition of the country.

The role language for this conversation is that a supply-chain manager designs or models the chain, breaks a messy network problem into a straight one, and makes the trade-off explicit. On a team that configures and optimizes the outbound network, that posture is the configuration column. The program manager does not pick every ZIP code. The program manager names the decision, the owner, the failure, and the check before and after.

Nothing in this table is an internal system name. "Who typically owns" is the generic role in a network of this shape.

DECISION	WHO TYPICALLY OWNS	WHAT BREAKS IF THE CONFIG IS WRONG	WHAT A PROGRAM MANAGER CHECKS BEFORE AND AFTER
Sort code and destination mapping (configuration)	Transportation configuration, with the sort building	Packages ride to the wrong delivery station or the wrong carrier. Rehandle, a missed day, and a second linehaul.	Before: a sample of ZIP codes against the building each one is supposed to reach. After: mis-sort rate and the cost of the packages that had to be sent again.
Ship method and customer cutoff (configuration)	The promise function, with transportation	The site offers a date the path cannot make, or it hides a date the path could make.	Before: the cutoff sits in front of the pull time, with the pack buffer still inside it. After: the share of units attempted on or before the promised date.
Critical pull time on a lane (configuration)	Fulfillment-center outbound and transportation together	An early pull time sends trucks out light. A late one leaves packages on the dock for the next cycle.	Before: the pull time against the transit-time distribution, not the average alone. After: share of volume that missed the pull time, and trailer fill on that lane.
Lane assignment, including a direct injection that skips the sort (configuration)	Network configuration	An extra touch on volume that could have gone direct, or a direct trailer that cannot fill and cannot make the window.	Before: touches and transit time against the promise. After: cost per package and the on-time share on that lane.
Delivery-station ZIP catchment (configuration)	Last-mile network design, with the station	Routes too long to finish inside the window, a station in overflow, or a neighboring station starved.	Before: drive time of the farthest ZIP, and packages per route. After: stops per hour, overtime, and rolled volume.
Volume forecast (planning)	Planning	Too few trucks, or labor and trailers paid for freight that does not show up.	Before: forecast against trailing actuals, including the peak week. After: rolled packages from no trailer, and cost per package from empty ones.
Labor plan at the sort center and the delivery station (planning)	Building operations	The sort window runs long, so the upstream pull time was fiction.	Before: planned throughput against the arrival curve. After: sort completion against the window.
Truck capacity and the carrier plan (planning)	Transportation planning	Freight left on the dock, or a trailer dispatched for a pile that never built.	Before: cube and weight against orders. After: fill, and the volume that rolled.
A new node type, or a new region boundary (design)	Network design, with finance	A building that does not fill, or a region that cannot hold the selection the promise assumes.	Before: demand, the in-region rate, and the cost-to-serve case. After: miles, touches, in-region fulfillment, and cost per unit.

> Owning a configuration change means owning the metric it will move. A pull-time change that helps on-time delivery and hurts fill is a successful change only if leadership accepted that trade before it shipped.

> **SAY THIS — 90 WORDS**
>
> A team like TESS lives in the configuration layer: the sort, the lane, the cutoff, the catchment, and the pull time that make a plan executable. Planning sets volume, labor, and trucks. Design changes the nodes and the region boundaries. A wrong configuration looks like a building problem and is usually a mapping problem. Before a change I would name the decision, the owner, and the metric that will move. After it I would read that metric against fill, touches, and the share of packages that still make the promise.

I.4 Reverse flow and the ordinary exceptions

The return path is the generic model, stated briefly because Amazon's public writing is about the forward promise. A customer return or an undeliverable package comes back to a station or a fulfillment center, is graded, and either returns to promiseable stock or leaves the network. It consumes the same doors, the same sort, and often a linehaul in the other direction. It does not repair a forward miss. The forward exceptions below are the ones that hit the promise on the day.

EXCEPTION	WHAT HAPPENS IN THE GENERIC MODEL	METRIC IT HITS
Late linehaul	The trailer arrives after the delivery station has started, or finished, building routes. Packages miss the van and roll to the next cycle unless the station holds the route, which pushes overtime onto every stop behind them.	On-time attempt against the promised date. Also route overtime if the station waits.
Missed critical pull time	The package is still in the fulfillment center when its trailer closes. The next departure is the recovery, and it may be a day later. A recovery truck or air can buy the promise back at a higher cost per package.	Pull-time compliance first, then the promised-date attempt if there is no recovery.

EXCEPTION	WHAT HAPPENS IN THE GENERIC MODEL	METRIC IT HITS
Mis-sort	The package is on the wrong trailer. It arrives at the wrong building and needs a second sort and a second linehaul. The original route is short a stop; the receiving building has a package it did not plan.	Mis-sort rate, rehandle cost, and the promised-date attempt when the extra leg misses the day.
Delivery-station overflow	More packages arrived than the station has routes, vans, or drivers to clear. The excess rolls, or it is pushed onto Flex or an extra route at a higher cost per package.	Rolled volume, cost per package on the overflow path, and the promised-date attempt for what rolled.
Damaged package	The unit is pulled from the path. The customer is re-promised or refunded. The original package does not get faster by being re-taped into the same truck.	A damage or concession rate: cost and customer trust. Speed is affected only when a replacement is shipped.

A 2024 MIT thesis describes Amazon's internal Delivery Estimate Accuracy metric as the percentage of items with a delivery attempt on or before the promised delivery date. That is the right family of metric for the first four rows. Amazon does not publish the rate, and this packet does not invent one. The thesis also names two fulfillment-center causes, a virtual-physical inventory mismatch and a missort, as large contributors to misses. That matches the table: a missort is a configuration and execution defect, and it shows up as a late or wrong attempt, not only as a building-quality score.

> **SAY THIS — 85 WORDS**
>
> Exceptions are how the promise fails in the middle of the path. A late linehaul and a missed pull time both miss the van. A mis-sort sends the package to the wrong building. Overflow means the station has more packages than routes. Damage takes the unit off the promise. Each one has a different owner and a different metric. I would separate a speed miss from a cost miss before proposing a fix, because adding a truck and adding a recovery path are different decisions.

Part 4 — How Amazon thinks about network design

II. How Amazon thinks about transportation network design

Three objectives are in tension, and Amazon's own writing names all three. Cost to serve has to come down. Inventory has to sit where the order will be placed. The customer has to get the unit on the date the site promised. Herrington, July 31, 2023, put the mechanism in one place: regionalizing cut miles and handoffs, the distance from sites to the customer was down 15 percent since the start of that year, middle-mile touchpoints were down 12 percent, and same-day was both the fastest path and one of the cheapest. Jassy's letters and the 2024 and 2025 earnings calls then reported the outcome: per-unit cost to serve down, and Prime speed up, in the same years. The design problem is to move all three, and to say which one gives when they cannot.

II.1 Minimize cost per package

In the generic model the cost of a package is the sum of handling, linehaul, sort, and last mile, divided by the packages that actually moved. Four levers change that sum.

Truck utilization is cube against weight. A trailer fills out on one of them first. Small parcels usually fill the cube while the payload still has room, so the useful denominator is usable cubic feet, not the weight rating. The 2023 sustainability report names truck fill, the share of truck volume used, as a lever Amazon actually pulled.

Consolidation against frequency is the same trailer counted in time. Two departures a day keep the pull time early and leave each trailer light. One departure fills the trailer and makes every package that missed the single cutoff wait until tomorrow's clock. Amazon Science (October 14, 2022) describes the operating version: a lane is one or more trucks across the day, and the plan assigns volume so the truck does not leave empty and the dock is not left holding a handful of packages.

Fewer touches is a bypass. A package that goes fulfillment center, sort center, delivery station is handled three times on the middle and last mile. A direct injection from the fulfillment center to the delivery station deletes the sort. Zone skipping, in the generic model, is the same idea aimed at a carrier: induct the package closer to the destination so you do not buy the carrier's long zone. Amazon Science (July 24, 2023) and the 2025 INFORMS paper both describe the direct fulfillment-center-to-delivery-station arc as a designed output of regionalization, because localized volume can fill a direct trailer.

Last-mile density is stops per hour, which shows up as packages per route. A van costs roughly the same whether it carries a light route or a full one. The catchment, the station's wave, and the promise together set how many stops fit before the window closes.

TRAILER FILL ON A 53-FOOT TRAILER

Assumptions (illustrative unless cited): usable cube is 3,000 cubic feet; an average package is 0.5 cubic feet and 2 pounds; payload room is 45,000 pounds, so this freight hits cube before weight; the lane costs $1,512 however full the trailer is; compare 70 percent cube fill with 90 percent.

STEP	FORMULA	SUBSTITUTION	RESULT
1. Cube at 100 percent	usable cube / cube per package	3,000 / 0.5	6,000 packages
2. Weight at that cube	packages x pounds	6,000 x 2	12,000 pounds
3. Which constraint binds	weight against payload	12,000 vs 45,000	Cube binds. Weight does not.
4. Packages at 70 percent	fill x cube capacity	0.70 x 6,000	4,200 packages
5. Cost per package at 70 percent	lane cost / packages	$1,512 / 4,200	$0.36
6. Packages at 90 percent	fill x cube capacity	0.90 x 6,000	5,400 packages
7. Cost per package at 90 percent	lane cost / packages	$1,512 / 5,400	$0.28
8. Savings	70 percent cost - 90 percent cost	$0.36 - $0.28	$0.08 per package, 22 percent lower linehaul

The trailer did not get cheaper. The same $1,512 was divided by 1,200 more packages. A dense, heavy SKU would flip step 3 and make weight the binding constraint. The check before a fill target is which one binds on that lane.

TWO LIGHT DEPARTURES, OR ONE FULL DEPARTURE

Assumptions (illustrative unless cited): a trailer has 1,000 package positions and costs $570 to run the lane; two departures a day at 60 percent fill move 1,200 packages; one departure at 95 percent fill moves 950 packages, so the policies are not the same daily volume; with departures spaced evenly, mean wait for the next truck is half the gap between trucks.

STEP	FORMULA	SUBSTITUTION	RESULT
1. Packages per truck, frequent	fill x positions	0.60 x 1,000	600 packages
2. Cost per package, frequent	truck cost / packages on that truck	$570 / 600	$0.95
3. Daily throughput, frequent	trucks x packages	2 x 600	1,200 packages, $1,140
4. Packages on the single truck	fill x positions	0.95 x 1,000	950 packages
5. Cost per package, consolidated	truck cost / packages	$570 / 950	$0.60
6. Cost gap	frequent - consolidated	$0.95 - $0.60	$0.35 per package
7. Mean wait, two departures	half the gap	half of 12 h	6 hours
8. Mean wait, one departure	half the gap	half of 24 h	12 hours
9. Delay the consolidation adds	one-departure wait - two-departure wait	12 - 6	6 hours of average wait

You buy $0.35 a package by accepting six more hours of average wait, and you move 950 packages that day instead of 1,200. If the lane's real demand is 950, the honest frequent policy is two half-empty trucks, which is worse than 60 percent fill. If the demand is 1,200, one trailer cannot hold it and the consolidated policy is not available. The comparison is a design choice about scheduled capacity, not a free saving.

PACKAGES ON THE ROUTE

Assumptions (illustrative unless cited): a route costs $264; there are 1.25 packages per stop; compare a route of 150 packages with a route of 220 packages; the route cost does not change with the count.

STEP	FORMULA	SUBSTITUTION	RESULT
1. Stops at 150 packages	packages / packages per stop	150 / 1.25	120 stops
2. Cost per package at 150	route cost / packages	$264 / 150	$1.76
3. Cost per stop at 150	route cost / stops	$264 / 120	$2.20
4. Stops at 220 packages	packages / packages per stop	220 / 1.25	176 stops
5. Cost per package at 220	route cost / packages	$264 / 220	$1.20
6. Cost per stop at 220	route cost / stops	$264 / 176	$1.50
7. Savings per package	thin route - dense route	$1.76 - $1.20	$0.56 per package

The denser route is cheaper per package and cheaper per stop because the same $264 covers 56 more stops. It is also longer. If those 56 stops push the last delivery past the promised window, the saving is real and the promise is broken. Density is a cost result that has to fit inside the pull-time chain from section I.2.

SAY THIS — 82 WORDS

Cost per package is every touch and every mile, divided by the packages that actually rode. The levers I would walk are trailer fill, how often a lane departs, whether a package can skip a sort, and how many stops a route can hold. Frequency buys a shorter wait and spends fill. Consolidation does the opposite. I would show the dollars per package and the hours of delay in the same sentence, so the trade is explicit before anyone picks a side.

II.2 Put inventory where it changes the ship

Regionalization is an inventory decision whose output is miles and touches. Jassy's 2022 shareholder letter, published in 2023, describes the prior state: one national U.S. network, so a miss at the local fulfillment center shipped from somewhere else in the country, at higher cost and a longer delivery. The replacement was eight interconnected regions, each with broad selection, largely self-sufficient, still able to ship nationally. The letter says the redesign had just been completed and that shorter distances mean a lower cost to serve and a faster order.

Amazon Science, July 24, 2023, gives the selection logic and the result. Eight was as many regions as the model would support before speed suffered or inventory had to move between regions so often that the design defeated itself. The Northeast and Mid-Atlantic regions switched on January 18, 2023, and the other six were pulled forward. Orders fulfilled entirely from fulfillment centers inside the region went from 62 percent before the switch to 76 percent. Herrington, July 31, 2023, stated the customer-facing version: more than 76 percent of fulfilled orders came from within the customer's region, distance from sites to the customer was down 15 percent from the start of the year, and middle-mile touchpoints were down 12 percent. The 2023

sustainability report adds the volume and the miles: in the fourth quarter of 2023 Amazon shipped nearly 544 million more items from in-region fulfillment centers than in the fourth quarter of 2022, and in-region shipping to delivery stations avoided nearly 16 million miles in 2023. The 2025 INFORMS paper dates full deployment to March 2023 and describes the method: cluster ZIP codes, assign fulfillment-center capacity to the clusters, simulate, then judge the design on cost to serve and delivery speed.

On the Q4 2025 earnings call, February 5, 2026, finance said the U.S. regionalized network was operating at scale and that the next productivity work was still inventory placement: less distance, fewer touches per package, better package consolidation. The 76 percent, the 15 percent, and the 12 percent are 2023 results. They were not restated as 2025 rates.

The inventory math under that story is the generic square-root law, associated with Maister (1976). It is not an Amazon formula. If demand is split across n equal, independent regions, and lead time and the service level stay the same, total safety stock grows with the square root of n. Pool the slow movers, where a long ship is rare. Spread the fast movers, where almost every unit would otherwise travel.

SAFETY STOCK AT ONE NODE AND AT EIGHT REGIONS

Assumptions (illustrative unless cited): one national demand stream with a weekly standard deviation of 400 units; cycle-service z of 1.65; lead time of 1 week; the eight regions are equal and independent, so each region's standard deviation is the national standard deviation divided by sqrt(8); safety stock is z times sigma times sqrt(lead time).

STEP	FORMULA	SUBSTITUTION	RESULT
1. Safety stock, one node	z x sigma x sqrt(L)	1.65 x 400 x sqrt(1)	660 units
2. Scale factor for 8 regions	sqrt(n)	sqrt(8) = 2.828	2.828 times
3. Safety stock, 8 regions	SS_1 x sqrt(n)	660 x 2.828	1,866 units
4. Exact value without the rounded root	660 x sqrt(8)	660 x 2.828427	1,867 units
5. Extra stock from splitting	SS_8 - SS_1	1,867 - 660	1,207 units
6. Check from the per-region sigma	n x z x (sigma / sqrt(n)) x sqrt(L)	8 x 1.65 x (400 / 2.8284)	1,867 units

Splitting this demand into eight regions holds about 2.83 times the safety stock, 1,207 more units, for the same cycle service and the same one-week lead time. That is the inventory bill for being local. It is worth paying when the transportation saved on the flow is larger than the holding cost of those 1,207 units, which is the fast-mover case below. It is not worth paying when the flow is thin.

EXPECTED MILES, ONE NATIONAL DRAW AGAINST EIGHT REGIONAL SOURCES

The distance table is illustrative. It is the mileage from a customer in region 4 to a fulfilling region. Under a national policy the fulfilling region is equally likely to be any of the eight. Under a regional policy it is region 4.

FULFILLING REGION	MILES TO THE CUSTOMER IN REGION 4	NATIONAL PROBABILITY
Region 1	1,800	1/8
Region 2	1,200	1/8
Region 3	600	1/8
Region 4, the customer's region	250	1/8
Region 5	600	1/8
Region 6	1,200	1/8
Region 7	1,800	1/8
Region 8	2,200	1/8

Assumptions (illustrative unless cited): the miles in the table above; national sourcing draws the fulfilling region with equal probability; regional sourcing always ships inside region 4 at 250 miles; this is not a reconstruction of Amazon's 15 percent result.

STEP	FORMULA	SUBSTITUTION	RESULT
1. Sum of source miles	add the eight distances	1,800 + 1,200 + 600 + 250 + 600 + 1,200 + 1,800 + 2,200	9,650 miles
2. Expected miles, national	sum / 8	9,650 / 8	1,206 miles

STEP	FORMULA	SUBSTITUTION	RESULT
3. Expected miles, regional	in-region distance	250	250 miles
4. Miles avoided in this drawing	national - regional	1,206 - 250	956 miles

Do not quote 956 miles, or a drop from 1,206 to 250, as Amazon's result. This drawing assumes the pre-change network shipped from a random region. Amazon Science said 62 percent of orders were already fulfilled inside the future region before the switch, so the real base was already partly local. The published result to cite is Herrington, July 31, 2023: distance from sites to the customer down 15 percent, and middle-mile touchpoints down 12 percent, since the start of that year.

ONE ASIN, SPREAD OR POOLED

Assumptions (illustrative unless cited): in-region ship costs $3 and cross-region ship costs $6; demand is spread evenly, so two stocked regions cover 25 percent of demand in-region and 75 percent cross-region; the fast mover sells 5,000 units a week and the extra safety stock of eight regions versus two is 1,130 units held at $0.07 per unit per week; the slow mover sells 20 units a week, and each region that stocks it carries a 15-unit minimum because the case pack dominates the square-root safety stock, held at $0.96 per unit per week.

STEP	FORMULA	SUBSTITUTION	RESULT
1. Fast mover, extra transport if held in 2 regions	cross-region units x premium	0.75 x 5,000 x ($6 - $3)	$15,000 per week
2. Fast mover, extra holding if placed in 8	extra units x hold rate	1,130 x $0.07	$79 per week
3. Fast-mover decision	transport saved - holding added	$15,000 - $79	Place in 8 regions. Net about $14,900 a week.
4. Slow mover, stock if placed in 8	regions x minimum	8 x 15	120 units
5. Slow mover, stock if held in 2	regions x minimum	2 x 15	30 units
6. Slow mover, extra holding to spread	extra units x hold rate	(120 - 30) x $0.96	$86 per week
7. Slow mover, extra transport if held in 2	cross-region units x premium	0.75 x 20 x ($6 - $3)	$45 per week
8. Slow-mover decision	holding added - transport saved	$86 - $45	Hold in 2 regions. Spreading costs $41 a week net.

The fast mover pays for its own forward placement: about $79 of holding avoids about $15,000 of long shipments. The slow mover does not. Spreading it costs $86 a week of inventory to avoid $45 of freight, so the cross-region ship is the cheaper policy, and those 15 long shipments a week are an accepted cost. The 2024 inbound placement fee is this comparison turned into a price the seller sees. Amazon's late-2023 announcement said the fee reflects the cost of distributing inventory to fulfillment centers close to customers, and that the seller can remove it by doing that distribution.

SAY THIS — 88 WORDS

Regionalization is an inventory decision that shows up as miles and touches. Fast movers earn a forward position in each region because the extra safety stock is small next to a long ship. Slow movers do not, and the cross-region ship is the cheaper plan. The square-root law is why safety stock grows as demand is split, and it is a generic inventory model, not an Amazon formula. I would place an ASIN by setting the holding cost of another region next to the freight that placement avoids.

II.3 The customer gets it on the day that was promised

A promise is a radius plus a clock. Same-day requires the unit inside the metro, in a building that can pick, sort, and dispatch before the afternoon route leaves. Herrington's 2023 description of the same-day building is that path. Next-day, or Prime one-day, requires the unit inside the region and a pull time that connects the fulfillment center, the sort, and the station inside one cycle. Two-day is the national backup when the region does not hold the unit. Jassy's 2024 letter, published in 2025, gives the selection that sits on those tiers: Prime began as two-day on a million items and had grown to more than 300 million items, with tens of millions available in one day or better, and a rising share same-day. The letter says the speed came from the regional redesign, new placement algorithms, and same-day fulfillment centers, and that a faster promise raises the rate at which customers complete the purchase and the rate at which they come back.

The cost curve is not "faster always costs more." Jassy's 2023 letter, published in 2024, called the same-day facilities the lowest cost to serve in the network, because they delete middle-mile distance. Herrington said the same thing in July 2023. The

expensive version of faster is the one that keeps the old path and buys frequency or air on top of it. The cheap version deletes a touch.

Reliability is a separate axis from the promised speed. The customer help page explains how the date is built. It does not publish the share of attempts that hit it. The 2024 MIT thesis names that share, internally, Delivery Estimate Accuracy: items attempted on or before the promised delivery date. Cite it as an internal metric described in a public thesis. Do not cite a percentage. On the Q4 2025 call, February 5, 2026, Amazon reported a third year of fastest-ever Prime speeds together with a lower cost to serve, and said U.S. Prime members received over 8 billion items the same or next day in 2025, up more than 30 percent year over year. Speed and the cost line moved together in the reported years. The accuracy rate was not part of that disclosure.

HOW FAR A STATION REACHES, AND HOW MANY A METRO NEEDS

Assumptions (illustrative unless cited): average speed is 45 miles per hour; the outbound window is 3 hours if the vehicle only drives; a 60-minute drive-time cap is the coverage rule; a station's coverage is a circle; the illustrative metro and its surroundings cover 4,000 square miles; that metro originates 60,000 packages a day, a route holds 200, and a station runs 50 routes in the wave; pi is 3.1416.

STEP	FORMULA	SUBSTITUTION	RESULT
1. Upper-bound radius	speed x window	45 mph x 3 h	135 miles
2. Radius under a 60-minute cap	speed x 1 hour	45 x 1	45 miles
3. Area of that circle	pi x radius squared	3.1416 x 45 x 45	6,362 square miles
4. Stations the drive time requires	metro area / circle area, rounded up	4,000 / 6,362	1 station. The circle is larger than the metro.
5. Routes the volume requires	daily packages / packages per route	60,000 / 200	300 routes
6. Stations the volume requires	routes / routes per station	300 / 50	6 stations
7. Which constraint binds	compare the two station counts	6 against 1	Volume binds. Drive time does not.

The 135 miles is an upper bound: it assumes three hours of driving and no stops. A real route spends most of its window at stops, so the practical radius is well inside 45 miles, and step 4 would rise. Even then, in this illustration the package count asks for six stations before the map does. Drive time sets a floor. Doors, vans, and the sort wave set the number you build. Using the 135-mile figure as a station spacing would under-build a metro.

WHAT ONE DAY FASTER COSTS

Assumptions (illustrative unless cited): the ground lane is the consolidated truck from section II.1, $570 for 950 packages, $0.60 per package, one departure a day; 30 percent of packages are packed after that departure and would otherwise be promised a day later; a second truck doubles the truck cost and splits the same 950 packages; air on a long lane costs $4.00 per package more than the ground it replaces, and it is used only on the 30 percent.

STEP	FORMULA	SUBSTITUTION	RESULT
1. Ground cost per package	truck cost / packages	$570 / 950	$0.60
2. Two trucks, same volume	(2 x truck cost) / packages	$1,140 / 950	$1.20
3. Blended increase from the second truck	two-truck cost - ground	$1.20 - $0.60	$0.60 on every package
4. Cost per package whose promise improved	blended increase / share improved	$0.60 / 0.30	$2.00 per package pulled forward
5. Fill on the two-truck plan	packages / (trucks x positions)	950 / (2 x 1,000)	47.5 percent
6. Air, blended across the lane	share flown x premium	0.30 x $4.00	$1.20 on every package
7. Compare	second truck against air	$0.60 vs $1.20 blended	The second truck is cheaper when a few hours of earlier departure is enough to gain the day. Air is the tool when the transit itself cannot make the day by road.

Paying $0.60 on every package to pull 30 percent of them forward is $2.00 per package you actually accelerated, and it cuts fill from 95 percent to 47.5 percent. Flying that 30 percent costs more in this illustration, $1.20 blended. The design choice follows the constraint. If the miss is a cutoff inside a region, add the departure or move the pull time. If the miss is distance, air or a closer inventory position is what changes the promise. Buying air on the whole lane, including packages that were already on time, is the version of this math with the share set to 100 percent: $4.00 instead of $1.20.

> **SAY THIS — 89 WORDS**
>
> A delivery promise is a radius and a clock. Same-day, next-day, and two-day are different drive times and different pull times, and each one has a cost. Delivery estimate accuracy, an internal metric described in a public thesis, is the share of units attempted on or before the promised date. Amazon has not published the rate. I would not buy a faster promise with a blanket air upgrade. I would price the units that miss the ground cutoff, then choose an earlier truck or air only for that slice.

II.4 How a network designer decomposes the problem

The decomposition Amazon has actually described, in the 2025 INFORMS paper and in Amazon Science (July 24, 2023 and October 14, 2022), lines up with the way an industrial engineer would expect a manager to hold the problem.

Define the demand surface. The paper treats five-digit ZIP codes as the points, on a surface that is dense in some corridors and thin in others. A national average is the wrong input. The Northeast and the Northwest do not cluster the same way, which is why the authors say a single algorithm was not enough and the boundaries were adjusted by hand.

Define the nodes and what each one costs. A fulfillment center costs labor, space, and a safety-stock position. A sort center or a hub costs a touch and a trailer. A delivery station costs routes. A same-day building costs a constrained assortment and deletes a middle-mile leg. Direct fulfillment-center-to-station lanes cost a trailer and save a sort. The cost function is lumpy: a truck costs nearly the same at 60 percent fill and at 95 percent, which is why fill shows up as cost per package.

Set the service constraint before the solve. Cost to serve and delivery speed were the success metrics the paper names for the regionalization pilot. A design that cuts cost and misses the promise fails the constraint. A design that makes the promise by emptying the trailers fails the other one.

Solve at the level a manager can say out loud. One new building wants to sit near the center of gravity of the demand it will serve, pulled toward the dense ZIP codes and away from cheap land that adds drive time. Several buildings sharing one market is a p-median idea: choose p sites so the demand-weighted distance, inside the service radius, is as small as it can be. Region boundaries are a clustering of ZIP codes under a cap on demand and on distance, which is the method the INFORMS paper reports, followed by an assignment of fulfillment centers to those regions. Lane frequency is an assignment of flow to paths, including the choice to skip the sort. None of these needs a formula in the conversation. Each one needs the input, the constraint, and the output.

Stress the answer on peak and on failure. Peak is a volume the average day does not see, and it breaks pull times, door counts, and route caps together. A node down, or a carrier late, is a missing path: the packages reroute, the fills change, and some promises move out by a day. Amazon Science describes the planning analogue of safety stock: buffer transportation capacity held against demand uncertainty, and a plan stable enough that the people who hired labor and booked trucks can actually run it. A design that only works on the mean, with no buffer and no second path, is not finished.

QUESTION	DATA NEEDED	MODEL	OUTPUT	DECISION OWNER
Where should the next building go?	Demand by ZIP, drive time, labor and land cost	Center of gravity for one site. A p-median view when several sites share the demand.	A short list of locations and the demand each one captures inside the radius.	Network design
How many regions, and where are the boundaries?	The demand surface, fulfillment-center capacity, how deep the selection must be	Cluster ZIP codes under a demand cap and a distance cap, then assign buildings. This is the approach Amazon published for the eight-region design.	Boundaries, and which buildings serve which region.	Network design, with finance on the cost case
Which lanes run, and how many trucks?	Forecast volume, cube, weight, transit time, the pull time	A linear program. Variables are trailers by lane and departure. Objective is cost. Constraints are cube, the pull time, and the promise.	A trailer plan, expected fill, and cost per package.	Transportation planning
What pull time is safe on a lane?	Transit-time history, sort duration, what a miss costs to recover	A simulation, or a chosen quantile of the transit distribution. The average is the wrong input.	A cutoff, and the share of trucks that miss it on a bad day.	The configuration owner, with the building

QUESTION	DATA NEEDED	MODEL	OUTPUT	DECISION OWNER
How many dock doors?	Trailers per hour, minutes to unload, how uneven the arrivals are	A queue. Watch utilization and the wait in the tail, because the tail is what misses the pull time.	A door count that keeps the tail inside the cutoff.	Building design and operations
Did the change survive peak and a failure?	Miles, touches, fill, on-time attempts, cost per package, plus one peak week and one missed linehaul	Compare the plan to the week after, and to the stress case. No new solver.	Keep the change, tune it, or roll it back.	The program manager

How to talk about this with an industrial engineer. Avinash's background is operations research, Lean Six Sigma, and production planning, so the useful altitude is the decision, the binding constraint, and the tail. A lane plan is a linear program: trailers by lane and by departure, cost as the objective, cube and the pull time as the constraints. You do not solve it in the room. You say whether fill or the clock is binding. A pull time is a quantile of a transit-time distribution, and a simulation is how you see the share of trucks that miss it, which the average hides. Dock doors are a queue: as utilization gets high, the wait grows out of proportion, and the packages in that wait miss the pull time. The sentence to offer is the binding constraint, the metric you would read the week after, and what you would roll back if the tail shows up. Leave the solver unnamed.

> **SAY THIS — 90 WORDS**
> I would decompose a network change the way an industrial engineer expects. Start with the demand surface, name the nodes and what each one costs, set the service constraint, then choose a location or a lane plan and stress it on peak and on a missed truck. A linear program assigns lanes, a simulation sets the pull time against a transit-time distribution, and a queue tells you when a dock will miss that pull time. I would talk about the binding constraint and the decision, and leave the solver unnamed.

II.5 Public facts cited

Numbers in the worked examples are illustrative unless the sentence and this list both name a source. Years below are the year of the fact, with the publication year where it differs.

1. Eight interconnected, largely self-sufficient U.S. regions, replacing one national fulfillment network. Andy Jassy, 2022 letter to shareholders, published 2023. https://www.aboutamazon.com/news/company-news/amazon-ceo-andy-jassy-2022-letter-to-shareholders
2. More than 76 percent of fulfilled orders came from within the customer's region; distance from sites to the customer down 15 percent since the start of 2023; middle-mile touchpoints down 12 percent; same-day buildings fulfill, sort, and deliver from one site; pick to outbound dock averages 11 minutes; same-day across more than 90 U.S. metro areas. Doug Herrington, July 31, 2023. https://www.aboutamazon.com/news/operations/doug-herrington-amazon-prime-delivery-speed
3. In-region fulfillment, measured as orders fulfilled entirely from fulfillment centers inside the region, rose from 62 percent to 76 percent. Pilot regions Northeast and Mid-Atlantic switched on January 18, 2023. Eight was the highest region count that preserved speed without excessive inter-regional inventory movement. Direct fulfillment-center-to-delivery-station trucks became easier to fill. Amazon Science, July 24, 2023. https://www.amazon.science/news-and-features/how-amazon-reworked-its-fulfillment-network-to-meet-customer-demand
4. Full regional deployment by March 2023. Region design described as ZIP-code clustering under demand and distance limits, then fulfillment-center assignment, then simulation, judged on cost to serve and delivery speed. Middle-mile buildings specialized into hubs, for long-haul fill, and last-mile sort centers, for delivery stations. INFORMS Journal on Applied Analytics, 2025. https://pubsonline.informs.org/doi/full/10.1287/inte.2025.0295
5. 2023 restructuring into eight regions. Fourth quarter of 2023: nearly 544 million more items shipped from in-region fulfillment centers than in the fourth quarter of 2022. In-region shipping to delivery stations avoided nearly 16 million miles in 2023. More than two-thirds of Amazon packages in 2023 were delivered by Amazon's own logistics network. Truck fill named as an operating lever. 2023 Amazon Sustainability Report, published 2024. https://sustainability.aboutamazon.com/2023-amazon-sustainability-report.pdf
6. In 2023, more than 7 billion items arrived same or next day, including more than 4 billion in the U.S. and more than 2 billion in Europe. Same-day or overnight items in the U.S. up nearly 70 percent year over year. First global per-unit cost-to-serve reduction since 2018. U.S. cost to serve down by more than $0.45 per unit year over year. Inbound placement called out as 2024 work. 58 same-day facilities, top 100,000 SKUs, millions of other SKUs injectable from nearby fulfillment centers, lowest cost to serve in the network. Andy Jassy, 2023 letter to shareholders, published 2024. https://www.aboutamazon.com/news/company-news/amazon-ceo-andy-jassy-2023-letter-to-shareholders

7. 2024 was the second year in a row of a lower global cost to serve per unit. Fourth-quarter transportation productivity came from better inventory placement, higher units per package, and shorter distances. CFO remarks, Q4 2024 earnings call, February 6, 2025. https://www.freightwaves.com/news/amazon-beats-q4-expectations-forecasts-slower-q1-growth
8. Prime selection above 300 million items, with tens of millions available in one day or better. Speed attributed to regionalization, placement algorithms, and same-day fulfillment centers. A faster promise raises purchase completion and purchase frequency. Rural plan: more than a billion packages a year across 13,000 ZIP codes and 1.2 million square miles. Andy Jassy, 2024 letter to shareholders, published 2025. https://www.aboutamazon.com/news/company-news/amazon-ceo-andy-jassy-2024-letter-to-shareholders
9. For the third year in a row, in 2025, fastest-ever Prime delivery speeds globally and a lower cost to serve. U.S. Prime members received over 8 billion items the same or next day, up more than 30 percent year over year. Next productivity levers named as distance, touches per package, and package consolidation. The regionalized U.S. network described as operating at scale. Q4 2025 earnings call, February 5, 2026. https://www.fool.com/earnings/call-transcripts/2026/02/05/amazon-amzn-q4-2025-earnings-call-transcript/
10. Same Day Fulfillment Centers, SSDs: more than 85 in the U.S., top 90,000 SKUs, more than 500 million same-day units in 2026 year to date at the time of the letter. More than $4 billion committed to the rural delivery network. Same-day fresh grocery in more than 2,300 towns and cities. Andy Jassy, 2025 letter to shareholders, published 2026. https://www.aboutamazon.com/news/company-news/amazon-ceo-andy-jassy-2025-letter-to-shareholders
11. NielsenIQ click-to-door of 1.9 days for Amazon versus 4.4 days for other U.S. digital retailers. The measurement reported by eMarketer is July 2022, from an August 2022 NielsenIQ analysis. eMarketer, January 11, 2023. https://www.emarketer.com/content/delivery-sped-up-us-retailers-amazon
12. The same 1.9-day figure, "as of July," NielsenIQ via Insider Intelligence. Business Insider, January 18, 2023. https://www.businessinsider.com/amazon-package-delivery-speed-walmart-target-2023-1
13. Amazon cited NielsenIQ in February 2023: packages arrived 1.9 days after the order on average, about half the time of other retailers. The post does not restate the July 2022 measurement window. Amazon, LinkedIn, February 24, 2023. https://www.linkedin.com/posts/amazon_recent-nielseniq-data-shows-that-on-average-activity-7034899766161330176-ytJc
14. Inbound placement service fee effective March 1, 2024, averaging $0.27 per unit for standard-size products and $1.58 for Large Bulky. Reason stated: placing inventory close to customers improves speed and lowers transportation cost. Fee reduced or removed when the seller ships to multiple locations. Outbound fulfillment fees cut by an average of $0.20 (standard-size) and $0.61 (Large Bulky) from April 15, 2024. Amazon Selling Partners announcement, late 2023. https://sellingpartners.aboutamazon.com/Update%20to%20US%20referral%20and%20Fulfillment%20by%20Amazon%20fees
15. Placement options tied to that fee: generally four or more inbound locations at no fee, two or three at a reduced fee, and a single location at the fee, with Amazon spreading the inventory. Amazon seller documentation, as posted in a Seller Central discussion of the March 1, 2024 fee. https://sellercentral.amazon.com/seller-forums/discussions/t/f9677779-2a5f-42f6-8ffa-252f902e556d
16. Collect and prepaid are distinct inbound purchase-order types. An Amazon reference number is mandatory for collect freight and is assigned to collect vendors. Collect and prepaid purchase orders are not booked on the same shipment line. The bill-of-lading field is what links the appointment to the advance ship notice. Appointments are requested and granted in Carrier Central. Amazon Carrier Central manual, public PDF, no publication year printed on the document. https://images-na.ssl-images-amazon.com/images/G/01/magicarp/common/Carrier_Central_Manual_USAmazon.pdf
17. SLAM expanded as scan, label, apply, manifest. Path described as fulfillment center, then sort center, then delivery station. Before the first-party network, deliveries were completed by carriers such as USPS and UPS. Lanes are air and ground. A lane is one or more scheduled trucks, filled so the truck does not leave empty and the dock is not left holding packages. Amazon Science, October 14, 2022. https://www.amazon.science/blog/maximizing-the-efficiency-of-amazons-own-delivery-networks
18. Fulfillment-center flow, including the SLAM step between pack and the outbound trailer, and the handoff to a sort center. Amazon fulfillment-center tour. https://www.aboutamazon.com/news/operations/amazon-fulfillment-center-photo-tour
19. Middle mile from fulfillment centers by freight and air into sort centers, sorted by ZIP, then to delivery stations or carrier partners such as the local post office. Relay described as the tool carriers use to book Amazon loads. Amazon, transportation network article. https://www.aboutamazon.com/news/operations/powering-amazons-transportation-network
20. Amazon Air's primary U.S. hub opened at Cincinnati/Northern Kentucky International Airport on August 11, 2021. Investment stated as $1.5 billion. Sortation building stated as 800,000 square feet, processing millions of packages per week. https://press.aboutamazon.com/2021/8/amazon-air-launches-state-of-the-art-air-cargo-hub-in-northern-kentucky-creating-more-than-2-000-jobs

21. Amazon Freight, in Relay for Shippers, is an inbound less-than-truckload and truckload option, including non-Amazon destinations in the lower 48. Distinguished on the same page from the partner-carrier program. https://supplychain.amazon.com/partner-carrier-program
22. Delivery Service Partner program: independent businesses delivering for Amazon. Page states the program launched in 2018. https://www.aboutamazon.com/impact/empowerment/delivery-partners
23. Last mile described as delivery stations plus Delivery Service Partners, Amazon Flex, and Hub Delivery. https://www.aboutamazon.com/news/transportation/amazon-investment-delivery-network-small-town-rural-us
24. Amazon Flex: drivers use their own vehicles; Amazon.com blocks are typically 3 to 6 hours; pickup is at a delivery station. https://flex.amazon.com/ and https://flex.amazon.com/support/faq
25. Delivery dates are shipping date plus transit time. An "order within" countdown is shown when it applies. Amazon Customer Service. https://www.amazon.com/gp/help/customer/display.html?nodeId=GV8H5D3MMAR7JBLF
26. Amazon relies on a limited number of shipping companies to deliver inventory inbound and completed orders to customers. Form 10-K for the year ended December 31, 2025. https://www.sec.gov/Archives/edgar/data/1018724/000101872426000004/amzn-20251231.htm
27. Critical pull time, in a 2016 MIT thesis on Amazon transshipment: the time an item must be picked, palletized, and on a truck. Order critical pull time: the time the package must depart Amazon, set from the ship-method cycle time. The same thesis describes the domestic cross-dock: vendor freight into the cross-dock, then transship to destination fulfillment centers. http://hdl.handle.net/1721.1/106075
28. Delivery Estimate Accuracy, in a 2024 MIT thesis: the percentage of items with a delivery attempt on or before the promised delivery date. The thesis discusses virtual-physical mismatch and missort as causes. No rate is published. https://hdl.handle.net/1721.1/155603
29. Inbound cross-dock as a facility type near ports and rail, feeding fulfillment centers, with sortation centers and delivery stations downstream. Academic description of Amazon's network, FHWA Talking Freight, December 2019, Jean-Paul Rodrigue. Not an Amazon statement. https://www.fhwa.dot.gov/Planning/freight_planning/talking_freight/december_2019/talkingfreight12_19_19jpr.pdf
30. Square-root law of safety stock: total safety stock scales with the square root of the number of locations when demand is split into equal independent pools at the same lead time and service level. Generic inventory model, Maister, 1976, "Centralisation of Inventories and the 'Square Root Law'," International Journal of Physical Distribution. No stable open URL was verified for this session.

LOOKED FOR, AND NOT USED AS AN AMAZON FACT

- An official aboutamazon definition, or any numeric standard, for critical pull time. The term is used as defined in the 2016 MIT thesis. The hours in the backward schedule are the generic model.
- A published Delivery Estimate Accuracy percentage, or any customer-facing on-time percentage for the U.S. network in 2023 through 2025.
- "Sub-same-day" as the official expansion of SSD. The 2025 shareholder letter expands SSD as Same Day Fulfillment Centers.
- A current official building type named "receive center," separate from the inbound cross-dock described in the 2016 thesis and the 2019 FHWA briefing.
- An official public list of the names of all eight regions. Amazon Science names the pilot pair, Northeast and Mid-Atlantic. The INFORMS paper discusses Northeast and Northwest as density extremes and shows an approximate map. The other names are not stated as a roster here.
- A dollar cost-to-serve reduction for 2024 or for 2025. The calls say the per-unit cost fell again. They do not restate a figure like the 2023 "more than $0.45."
- A current mix of packages among Delivery Service Partners, Amazon Flex, USPS, and UPS. The sourced facts are "more than two-thirds by Amazon's own logistics" in 2023, historical use of USPS and UPS, sort centers handing to the local post office, and the 2025 10-K reliance on shipping companies.
- Any internal system name for the transportation-engineering stack. Public papers name modeling approaches. Those names are not repeated here, so they do not get recited as inside knowledge.

SAY THIS — 85 WORDS

When I use a number I will give the year and the source, and I will say when a figure is Amazon's and when it is an illustration. The public record supports the eight-region shift, the in-region fulfillment rate, the distance and touch reductions, the placement fee, and three years of lower cost to serve alongside faster Prime speeds. It does not publish a critical-pull-time standard or a delivery-estimate-accuracy rate. Those stay in the generic model, and I would label them that way out loud.

Part 5 — Questions for Avinash, run of show, numbers card

6. Questions for Avinash and the TESS team

Ask fewer if he is talking. The ones to protect are the change path, the metric he is behind on, the ninety-day object, and the closer. Each question is something his background makes fair: he planned production across five sites, he has been at Amazon since 2017, and he talks about cost-benefit, models, and the limitation of a product.

A. The team's charter, and where it sits

1. Where does this team sit between the transportation program managers, the product managers, and the engineers in TITANS? Who writes the requirement, who owns the live config, and who can stop a change?

   Why: my seat has been the definition and the gate, without owning the system that executes it. A good answer names three owners, and a change an L5 cannot approve alone.

2. Is an L5 here a customer of the config software, a partner who writes its acceptance bar, or both?

   Why: at Flexport I wrote the requirements and the parity gate, and I did not lay the pipe. A good answer says which of those he expects in the first two quarters.

B. How a network design change goes live

3. Walk me through one change you have shipped, a lane, a critical pull time, or a sort pattern, from the request to the live config. Where are intake, validation, deployment, and rollback, and which of those is thin today?

   Why: I do not know his path, and I do not want to invent one. A good answer has a stage where a change is held, and a rollback that is a number or a reason it has not had to fire.

4. When a config is wrong, what actually breaks first: a missed pull time, a sort that cannot take the volume, or freight left at the building? How fast do you see it?

   Why: my sensor was a ticket. I need his sensor. A good answer names the break that costs the most and the clock on which someone owns it.

C. The metrics, and which one he is behind on

5. Of cost per package, on-time delivery, speed, and config error rate, which one is this team most behind on right now, and which of the others are you willing to trade this quarter to move it?

   Why: I will not optimize a metric he is not behind on. A good answer is one metric, and the trade he will actually sign.

6. How do you decompose cost to serve per package, and which term is a config choice rather than a rate you cannot touch?

   Why: sixty-six cents was useless until it split into tickets and platform. A good answer is a split, and one term an L5 can move by changing a config.

D. The first ninety days

7. What does an L5 on this team own in the first ninety days: a metric, a set of lanes, a config domain? What does day ninety look like that is not a tour of the network?

   Why: I want a countable object. A good answer is one domain, one metric, and one review I am expected to run.

E. Automation and models

8. Where is a model already choosing a route, a pull time, or a sort plan, and where is a person still the config because you already know how the model is wrong?

   Why: he has said he wants the limitation of a product early. That is the human loop I kept until a source held its rate. A good answer names the limitation and the re-entry rule.

9. When a model recommendation and a planner disagree, who wins, and what evidence reopens the decision?

   Why: my release rule was two-sided and signed. The formula did not silently re-plan a lane. A good answer is a written rule, not "the model is usually right."

F. How success is judged

10. How is this team judged at OP1 and OP2: the metric, the narrative, and the miss that still gets an L5 a hard question in that room?

Why: I have fed a leadership readout from a weekly review, and the grain of that room is the grain I will run to. A good answer is the one number that shows up, and an example of a miss that was well received because it was early.

The question to end on

11. If I were here in six months, what would you want me to have made boring, a config change, a cost line, or an exception, that is not boring today?

    Why: it asks for the real pain in his words, and it gives you something to close on. A good answer is one concrete object, not a list of values.

7. Sixty-minute run of show

MINUTES	WHAT YOU DO	WHAT YOU HOLD
0–5	Thank him. Sixty-second introduction, section 1. Stop. If he talks, listen. One sentence on the role you think this is: config, cost, and the clock.	The thirty-second cut, if he starts with his own introduction and leaves you little time.
5–10	Why this team, section 2, sixty seconds. Then ask question 3, the path from request to live config, and let him answer. That answer tells you which story to tell.	The public facts, unless the eight regions or the 1.9 days fit in one clause. Do not recite them as a briefing.
10–22	Analytical, first. Story 4.1, the lanes, the tail, the two-sided release. This is a transportation team. If he leans into cost, bridge to 4.3 and the cents, and be ready for the worksheet.	Story 4.2, the eight domains, unless he asks how the data got clean. Do not add the two dollars.
22–32	Cross-functional. Story 4.3b first if he has asked anything about comparing systems, deprecating a product, or changing a leadership decision: Crux at 63 against 4PL at 25, T-30 scored against T-0, 27 against 25, the residual you named, 4PL kept. Otherwise 4.3: gates, the held wave, cost per container. If he is more interested in a roadmap decision than a metric, tell 4.5 and say the five million once.	The automation percentage. The minutes line. The bulk price you do not have. Calling the 63 "a competitor's" after you have called it Crux.
32–40	Ownership. Story 4.4, the review you took because nobody owned it, twenty-four hours a week out of the priced baseline, parity within one percent. If he asks for a failure, switch to 4.10 and stop on the kickoff question.	4.6 and 4.7 unless the clock is healthy and he asks for a shorter process story. 4.8 and 4.9 only if he asks what you do now.
40–55	Your questions. Order if time is short: 3 the change path, 5 the metric he is behind on, 7 the ninety-day object, 8 where the model is already wrong, then the closer. Skip a question he has already answered.	Questions 1, 2, 4, 6, 9, and 10 if the conversation has already covered them. Do not read the list.
55–60	Close. One sentence: you configure a network so the package moves at the lowest cost that still meets the promise, and you want a gate in front of the change. Ask the closer if you have not. Stop. Thank him.	A new story. Do not open 4.11 in the last five minutes unless he asked how you would think about a promise network.

What to drop if the clock slips. Drop 4.7, then 4.6, then the dollar build-up inside 4.5. Keep 4.1 and 4.3. If you have told 4.3, you may skip 4.4 unless he asks how the review was built. Sephora stays down unless he asks for the current job. Design cases stay down unless he asks how you would think about a network you have not operated. A failure question replaces 4.7; it does not get added on top of a full ownership story. If you are at minute 35 and you have not left fifteen minutes for his answers and your questions, stop the story you are in and ask question 5.

Numbers card

Page is the count of form feeds in the packet text before the match, plus one. Status is how you may say it. "Record" means the reconciled career fact. "User anchor" means the packet carries it and the reconciliation says confirm it. "Illustrative" means a worksheet input chosen to land on a real endpoint, or a design-case model.

METRIC	VALUE	STORY	AMAZON PACKET PAGE	STATUS
Tenure	Fifteen years	Introduction	35	record
Manhattan peak	3 DCs, 100 stores; 99 percent fill vs 95 percent goal	Introduction only	34	record
Elementum years	2017–2019	4.1, 4.2, 4.7	717, 1240	record
Lanes measured	200 lanes; example brochure 18 days, mean 21, p99 34; about 400 shipments	4.1	472	record for the 200 and the 99 percent; example lane is the worked case
Lane A buffer	12,937 to 8,102 units; 4,835 out	4.1	473	illustrative lane; arithmetic matches the packet
Lead-time variance share	16,000,000 / 16,405,000 = 97.5 percent	4.1	473	illustrative lane
Program release	580,000 units, about $1.5 million, one time, 99 percent accuracy	4.1	474	record. Implied $2.59 a unit is derived. Do not add to the $2 million.
Lane B	5,238 to 6,621 units	4.1	474	illustrative second lane
Roll-up split	150 x 4,500 minus 50 x 1,900 = 580,000	4.1	474	illustrative split that ties to the record net
Spend cut	597 lanes = 80 percent of spend; remainder 20 percent	4.1	776	user anchor. Say "in my earlier record."
Completeness	60 percent to 95 percent; gate at or above 95 percent valid ASN	4.1	776	record. The 480/800 and 760/800 volumes are illustrative.
Freight on onboarded lanes	about 10 percent, about $2 million in six months	4.1, only if asked	777	user anchor. Implied $20 million base is derived. Different noun from the other two million.
Class-A accuracy	70 percent to 85 percent = 1 - MAPE of 0.30 then 0.15	4.2	444	record endpoints. The four-week 18.3 percent sample is illustrative and does not reproduce them.
Safety stock, forecast rule	about $2 million, one time, both errors improved	4.2	443	record. Never add to $1.5 million.
Alert	days of cover 1,200/100 = 12, against p90 of 14; live in 3 months	4.1 and 4.2	444	rule is record design; the 1,200, 100, and 14 are illustrative
Toolkit safety stock	2.33 x 100 x 4 = 932; at sigma 2, 466	4.2	443	illustrative
Eight domains	supplier, material, bill of materials, inventory, production, transportation, sales order, planning	4.2	1125	record. Manufacturer's domains. Contract manufacturer supplied the feed.
Customers	7 to 68 = 9.7 times; two months early	4.3	256	record. Do not round to ten.
Cost per container, endpoints	$0.66 to $0.03; about 95 percent	4.3	1331	record endpoints
Before worksheet	2,000 a week; 60 percent automated; 800 tickets; $1.50 and $120 illustrative; (1,200+120)/2,000 = $0.66	4.3	1331	illustrative rates chosen to land on the record endpoint. Do not say the minutes line. 12/60 x 75 = $15, not $1.50.
After worksheet	10,000 a week; 97 percent; 300 tickets; $0.80 and $150 illustrative; sum $0.039	4.3	1331	illustrative. Does not reproduce $0.03. Gap $0.009. Bulk price not on the page.

METRIC	VALUE	STORY	AMAZON PACKET PAGE	STATUS
Touch cousin check	$0.18 an eliminated interaction; 0.66/0.18 about 3.7; 0.03/0.18 about 0.17	4.3, 4.3b	1237; LSA WBR 2022 Forwarding Platform lines 2 and 12: 44,074.08 / 244,856 = 0.18	Finance's rate for one interaction, confirmed by the uploaded review sheet. Not the ticket price.
Crux against 4PL, raw	25 percent against 63 percent schedule reliability	4.3b	229, 1252	record. Your record also words the 63 as "a competitor's." Say Crux consistently.
Crux against 4PL, normalized	about 27 percent against 25 percent, same containers, sources, week, horizon	4.3b	229, 1252	record endpoints. The 1,000 / 400 / 252 / 270 / 250 populations are illustrative.
Residual gap	arrival variance 16 days against 8; six months of source additions	4.3b	229, 1252	record
Reliability after sources	90 percent against an 80 percent north star	4.3b	229, 1252	record
Scoring horizons	accuracy at ATD-28 and ATD-10; MAE; standard deviation of MAE at T-28, T-10, T-5	4.3b	LSA WBR 2022, Ocean FCL lines 30 to 33	record-from-docs. Say T-30 as the round number.
Crux base	79 customers; $15,491 a month = $185,892 a year; 294,581 containers in 2021; about 75 percent non-trade revenue	4.3b	61, 920	record-from-docs, RAPID. Cost to run Crux not on record.
Crux cancellations	August 1, September 1, October 24, 2022	4.3b	61	record-from-docs. No single switch-off date on record.
E-Team frame	February 1, 2022: Goal 1 lower cost to serve; measure CTS for 3PL plus 4PL against 4PL only; visibility team publishes its own review, arrival accuracy and interaction elimination first	4.3b, 4.4	uploaded E-Team takeaways; packet 16474	record-from-docs
FFA scope	905.9 engineer-months total, 75.5 engineer-years; P0 Export Partners FCL 126.9 engineer-months = 14 percent; P1 219.4; +15 percent pre-launch testing	4.5	uploaded FFA Options Summary	record-from-docs
Automation, spoken if asked	about 40 percent to about 90 percent	4.3	reconciled against the packet's 60 to 97 worksheet on 1331	record is 40 to 90. Worksheet is 60 to 97. Do not blend.
System accuracy	60 percent to 90 percent; parity within 1 percent	4.3, 4.4	270	record. Not arrival accuracy.
Wave hold	arrival accuracy 76 percent vs 80 percent	4.3	1331	record
Graduation gates	SLA 90, data quality 90, arrival 80, 100 containers, stable, satisfaction	4.3	256	record
Review hours	30 to 6 hours a week; 24 hours saved; $103,680 priced at an assumed $90	4.4	1331	record-from-docs arithmetic. Priced, not saved. $90 is assumed.
Snapshot grain	1.5 to 2 hours each; separately, about 4 hours a week at its worst	4.4	331	record, two grains. Do not stack them onto the 30 hours.
Security miss	two weeks; no Monday skipped	4.10	1237	record. No dollar on record.
Roadmap	month 18 to month 6; 66 percent less effort; about $5 million avoided once	4.5	376	record. Avoidance, not annual.
Model under the $5 million	12 x 18 x 66 percent x $20,000 = $2,851,200; plus $1.52 million pulled forward; $4.4 million vs $5 million, 12 percent under	4.5	376	modeled. Twelve engineers, $20,000, and 20,000 shipments are not measured counts.
Unit baseline	339.5 minutes; about $153 a shipment; 2,200 interactions	4.5	375	record inputs. The $0.45 blend is derived.
Import launch	April 2022; 11 partners, 10 countries; 48 of 53 import partners	4.5	375	record. Not the export launch.

METRIC	VALUE	STORY	AMAZON PACKET PAGE	STATUS
Carrier bid	about 6 weeks to about 3; gate under 5 percent	4.6	111	user anchor. No award dollar.
Austria quality	40 percent reported, about 55 percent true, about 99 percent within a quarter	4.7	223	record. Do not choose two months or three months.
Austria dollars	about $500,000 a year; about $8 million across 25 sites	4.7	223	record, executed only. The $5 million base is a user anchor. $320,000 a site is derived. Do not say a lane count.
Austria hours	8 to 10 hours a week before	4.7	223	record. The 1 to 2 hours after is a user anchor in the reconciliation.
Available-for-sale	signed June 5, 2025; oversell recurring to zero	4.8	167	record for the outcome. No baseline incident count. In-transit sigma 0.3 day is a user anchor, page 546.
Council dates	March 29, April 30, May 15, July 2, July 31, August 14; go-lives January and May 2025	4.9	627	record
Log and efficiency	714/836 = 85.4 percent; efficiency +15 percent; zero customer incidents	4.9	124	record. The 15 points have no before/after rates on record.
Robotics gate	98 percent bar; week two 94 percent; about 30 percent of volume held back; full in week three	4.9	598	record
Walmart promise and Google Fiber spine	no production figure	4.11	206 and 211	design case. Illustrative model only. Not delivered.

Public facts cited

- Amazon, February 24, 2023, citing NielsenIQ: about 1.9 days from order to delivery. The measurement window in the trade press is about July 2022. https://www.linkedin.com/posts/amazon_recent-nielseniq-data-shows-that-on-average-activity-7034899766161330176-ytJc
- Same figure, reported January 11, 2023, as NielsenIQ data. https://www.emarketer.com/content/delivery-sped-up-us-retailers-amazon
- Amazon, 2023, eight interconnected US regions. https://www.aboutamazon.com/news/operations/doug-herrington-amazon-prime-delivery-speed
- Same regionalization, from the 2023 shareholder letter as reported. https://www.supplychaindive.com/news/amazon-shifts-regional-fulfillment-model-faster-prime-delivery/647708/

> Those pages also print other operating figures, including an in-region fulfillment share and a middle-mile touchpoint change. They are public. They are not in the spoken blocks above. Use one only if he asks and you have the page, and label the year. Do not mix them into a career metric.